
Pancreatic Cancer LLM Clinical Trial System: From IND to Protocol to Legislation, Funding, and AI Peer Review

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Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Disclaimer: This work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 5.

The paper (v1.0) is at <https://doi.org/10.5281/zenodo.xxxxxxxx> and the repository (v4.6.0) is at <https://github.com/kevinkawchak/physical-ai-oncology-trials>. The paper files are deposited in [new-trial-system](#).

Keywords Pancreatic ductal adenocarcinoma; Investigational new drug; Clinical trial protocol; Large language model; AI peer review; Robotic pancreaticoduodenectomy; Daraxonrasib; Verification validation and uncertainty quantification; Small business innovation research; Physical AI

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Abstract

The prior pancreatic cancer trial system built its documents on a month scale, through teams, with peer review arriving after completion. The new system produces the same documents on a 1 to 4 day project time scale, through one author directing a master prompt, with AI peer review arriving during development. This paper discloses that system across four deliverables. The IND assembles twelve modules under 21 CFR 312 in four days. Two protocols follow: a Phase 1 first-in-human study at $n = 18$ using a 3+3 escalation behind a Phase 0 gate of at least 1000 simulations across two frameworks, and a Phase 2 randomized study at $n = 220$ across eight centers targeting a hazard ratio of 0.60. Five bill versions and four companion documents were deposited in eleven days. Fourteen funding artifacts followed, including a \$1,606,000 small-business ask against a \$3,500,000 five-year direct program, and a virtual trial projected at \$36,330 against an industry benchmark above \$120,000. Human peer review cannot absorb this output. A best-case round is seven to eight weeks and a typical one several months, against three model manufacturers reporting the same day; and review arriving after release cannot correct the released object. The two systems are incompatible, and the comprehensive provenance record the new one leaves is why it is trusted.

How to Read This Document

This paper is a disclosure of a working method, and its evidence is a set of artifacts that already exist and are already deposited. It is written so that four different readers can each open it at a different page and find the thing they came for within two pages. Table 1 names the four and the section each should open first. No reader is asked to read the document front to back, and every entry point named in Table 1 is a table or a figure rather than a paragraph.

Reader	Open at	Then	The question this reader is actually asking
Funding officer	§6, Table 18	§2, Figure 3	What was produced, at what cost, and by what repeatable method?
Regulatory reviewer	§3, Table 9	§4, Figure 10	Does the dossier map onto the codified content items, and where?
Legislative staffer	§5, Table 14	§5, Figure 14	What duty does the bill create, and who is positioned to discharge it?
Peer reviewer	§7, Table 21	§7, Figure 24	How was this reviewed, and what happened when the reviewers disagreed?

Table 1. Four readers, the section each should open first, and the one question each is actually asking of this paper's method and record.

Two indexes follow, and they are the paper's other entry points. Table 2 lists the twenty-five figures with the platform each is drawn on and the one question each answers, and Table 3 does the same for the twenty-five tables. A reader who knows the question but not the page should start at whichever of the two carries it. The figure count per platform is set by purpose and not by quota: six mermaid figures because six claims are chronological, six d2 figures because six are grids or containers, five graphviz because five are structural, four plantuml because four need a guard or a fork to be stated honestly, and four clustered infrastructure figures because four are about a system's parts.

Fig	§	Platform	Source directory	The question it answers
1	1	Mermaid	mermaid/	Which capability do eleven Federal actions leave unfilled?
2	1	D2	d2/	On which ten axes do the prior and new systems differ?
3	2	PlantUML	plantuml/	Which part of the build is concurrent, and which strictly serial?
4	2	Mermaid	mermaid/	What happens between one prompt and one pushed commit?
5	2	Graphviz	graphviz/	What does a figure specification store that a raster cannot?
6	3	Diagrams	diagrams-python/	Which source clusters fed which IND module?
7	3	Mermaid	mermaid/	How long did twelve IND modules take, against the prior system?
8	3	D2	d2/	Which file satisfies each codified IND content item?
9	3	Graphviz	graphviz/	What combination of failures would produce a clinical hold?
10	4	PlantUML	plantuml/	What guard fires each transition in a participant's state machine?
11	4	Mermaid	mermaid/	What must hold on each rung from Phase 0 to Phase 2?
12	4	D2	d2/	What did Phase 2 carry, replace, and add?
13	4	Diagrams	diagrams-python/	What is inside the site boundary, and what is outside it?
14	5	PlantUML	plantuml/	Which actor owns each statutory duty?
15	5	Graphviz	graphviz/	What did each bill version add that its predecessor lacked?
16	5	D2	d2/	How does one requirement reach the operating room?
17	6	Mermaid	mermaid/	When did each funding artifact appear, and how fast?
18	6	D2	d2/	What does the same direct work cost under two overhead regimes?
19	6	Graphviz	graphviz/	Where does each dollar land, and what does the award not reach?
20	6	Diagrams	diagrams-python/	What machinery turned one evidence store into fourteen artifacts?
21	7	Mermaid	mermaid/	How long does one review round take under each regime?
22	7	D2	d2/	What is the ratio between the two regimes on six economic axes?
23	7	PlantUML	plantuml/	Who reviews concurrently, and where does a human gate sit?
24	7	Graphviz	graphviz/	What happens when three model reviewers disagree?
25	8	Diagrams	diagrams-python/	What exposures remain, and what mitigates each?

Table 2. The twenty-five figures, their section, their platform, the directory carrying each one, and the question that each of them answers.

Tab	§	Contents	The question it answers
1	0	Four readers, four entry points	Which section should this reader open first?
2	0	Figure index, twenty-five rows	Where is each figure and what does it answer?
3	0	Table index, twenty-five rows	Where is each table and what does it answer?
4	1	Federal AI and cancer actions, Jan 2025 to Jul 2026	What has the government already committed to?
5	1	Prior system against new system, ten axes	What exactly changed, measured?
6	2	Master prompt and sub-prompt counts across works	How many times has this method been run?
7	2	Machine-readable diagram storage and reuse	What does the specification format buy?
8	3	IND package inventory by module	What is in the dossier?
9	3	IND section, source, and generation window	Where did each module come from, and how fast?
10	3	IND complexity and volume record	What scale did one author reach?
11	4	Phase 1 protocol synopsis	What is the first-in-human study?
12	4	Phase 2 protocol synopsis and deltas	What changed when it randomized?
13	4	Protocol generation metrics, both protocols	How long did two protocols take?
14	5	Statutory sections and the duty each creates	What does the bill require of whom?
15	5	Bill version lineage, v1.0 to v5.0	What did each version add?
16	5	Legislative artifact metrics	How much statutory text, how fast?
17	6	Ten funding applications and mechanism class	Who was asked, and under what authority?
18	6	Capitalization numbers	What is the ask, and what does it buy?
19	6	RFA-RM-27-001 v2 structure	What does a full NIH application contain?
20	6	Cost and time benchmarks, prior against new	What does the same work cost each way?
21	7	Prior against new peer review, nine measured axes	What does one review round cost each way?
22	7	Model roster and review role	Which model does what, and for whom?
23	7	AI peer review quality and efficiency metrics	How was review quality measured?
24	7	AI peer review instances across author works	How often has this been done before?
25	8	Limitations register with mitigations	What is still exposed, and what reduces it?

Table 3. The twenty-five tables, their section, their contents, and the one question each answers, so a reader who knows the question can find the page.

How This Paper Was Built

One master prompt was supplied by the author on August 12, 2026. Claude Code Opus 5 decomposed it into eight sub-prompts and executed them in order without further author intervention, committing and pushing every generated file the moment it was finished. A second author prompt on August 13, 2026 produced this update stage. The whole document, its twenty-five figures and its twenty-five tables were therefore finished within three days. The eight stages are recorded in new-trial-system/sub-prompts, and both prompts are filed verbatim in new-trial-system/prompts.

Stages 1 through 5 produced twenty-five machine-readable figure specifications, one file per figure, across five diagram platforms. Each specification carries a perspective statement naming what no other figure in the paper shows, the figure's two-line caption exactly as printed here, valid source in the platform's own syntax, a TikZ construction table of absolute coordinates with a stated node pitch, an edge-routing paragraph naming every edge that could cross a node and the clearance that prevents it, a value table attributing every number in the figure to a repository file, and a list of the exact sources read. No figure in this paper was drawn from memory of another figure, and no figure is a raster.

Stages 6, 7 and 8 produced the paper three times. The draft fixed the section set, the figure numbering and the table numbering, and carried a bracketed instruction in every section naming the repository file the next stage had to read. The full version drew all twenty-five figures, populated all twenty-five tables, discharged every instruction, and wrote the prose from the four author-final publication/ directories and the four input archives with direct quotation. The final version was the senior author's proof-reading pass. This update stage, deposited at new-trial-system/final-new-trial/update-final, repaired three `\foreach` headers that had stopped the final stage compiling, redrew Figure 14 as a class diagram, and rewrote §7 around the quantities of Table 21.

Each stage produced a self-contained Overleaf bundle, and each bundle is deposited beside its source. A reader who wants to check the production-rate claim of §2 against the record can read the branch history rather than this paragraph.

1 Introduction

Between January 2025 and July 2026 the executive branch committed to artificial intelligence and to cancer at the same time, and often in the same document. Executive Order 14179 removed barriers to American leadership in AI [1], and Executive Order 14212 established the MAHA Commission [2]. Executive Order 14355, signed September 30, 2025, is the only executive order aimed squarely at both topics: it tasks the MAHA Commission, in coordination with HHS, the Assistant to the President for Science and Technology, and the Special Advisor for AI and Crypto, with developing ways to use AI for improved diagnoses, treatments, cures, and prevention of pediatric cancer, beginning with the Childhood Cancer Data Initiative and with predictive modeling of patient response, disease progression, and treatment toxicity [3, 4, 5]. The same day, HHS raised the Initiative’s budget from \$50 million to \$100 million and committed to bringing private-sector partners in to apply advanced AI [6]. The MAHA Strategy of September 9, 2025 had already directed NIH and OSTP to develop an AI-driven approach to transform pediatric cancer research and clinical trials [7].

The machinery followed the mandate. Executive Order 14363 launched the Genesis Mission on November 24, 2025 [8, 9], and its NIH arm, the Bio Genesis Mission announced July 22, 2026, carries more than \$1.2 billion in obligated FY2026 and planned FY2027 funding across predicting the behavior of living systems, accelerating drug translation, and unlocking new pediatric cancer treatments [10, 11]. Inside the agencies, the FDA fielded Elsa on June 2, 2025, a generative tool to help staff synthesize information for tasks including clinical protocol review, and followed it with an agency-wide agentic platform on December 1, 2025 [12, 13]. CMS announced the WISeR model and, at the July 30, 2025 health technology event, an interoperability framework that was later institutionalized in a dedicated technology office [14, 15]. On June 22, 2026 HHS launched a department-wide effort to restore American leadership in clinical trials, of which Operation TrialBlazer, NCI working with cancer centers to streamline trial activation and improve enrollment, is the oncology arm [16, 17]. Table 4 places ten of these actions on one chronology with what each supplies to a sponsor.

Action	Date	What it supplies to a pancreatic cancer sponsor
EO 14179, Removing Barriers to American Leadership in AI	Jan 23, 2025	Policy cover for AI inside a regulated submission
EO 14212, MAHA Commission	Feb 13, 2025	The parent authority EO 14355 cites
FDA Elsa generative AI tool	Jun 2, 2025	An agency that already reads AI-assisted protocol text
MAHA Strategy	Sep 9, 2025	A direction to transform trials with AI
EO 14355, Unlocking Cures for Pediatric Cancer with AI	Sep 30, 2025	The only order aimed at AI and cancer together
CCDI budget doubled to \$100 million	Sep 30, 2025	Money for AI-driven oncology data work
EO 14363, Launching the Genesis Mission	Nov 24, 2025	An integrated Federal AI platform
HHS AI Strategy and the agentic platform	Dec 1 to 4, 2025	Agency-side capacity to process AI-assisted filings
HHS clinical trial effort and Operation TrialBlazer	Jun 22, 2026	Faster trial activation at cancer centers
Bio Genesis Mission	Jul 22, 2026	Over \$1.2 billion toward AI in biology

Table 4. Ten Federal actions between January 2025 and July 2026, and what each supplies to a sponsor preparing a pancreatic cancer trial dossier.

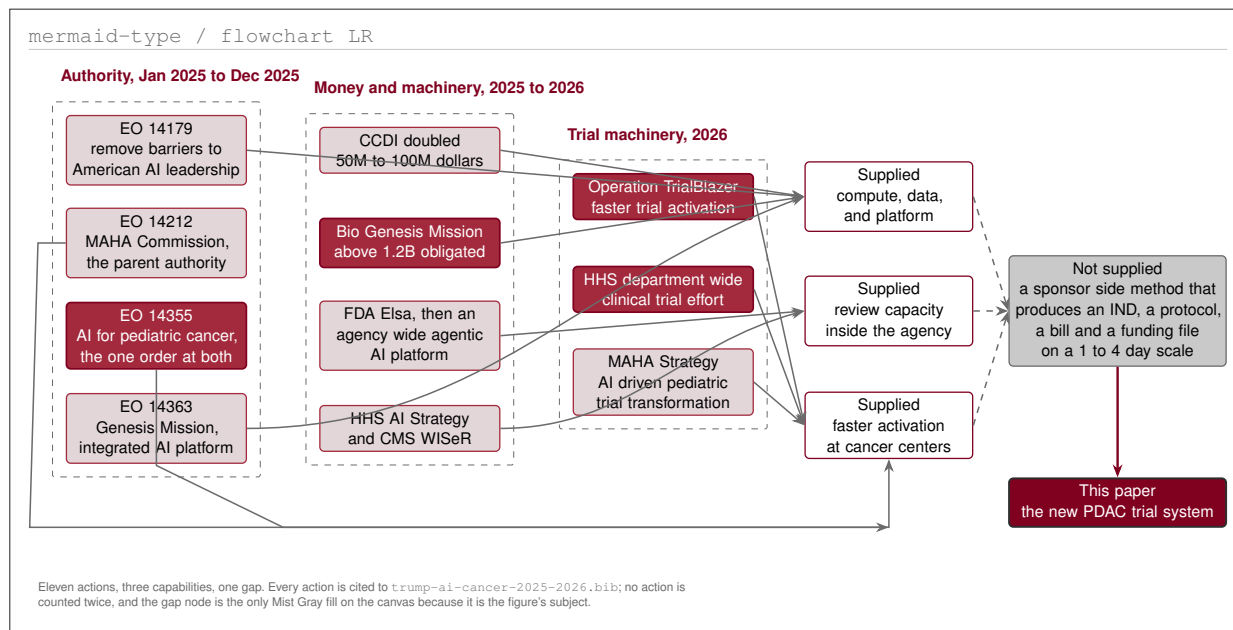


Figure 1. Eleven Federal AI and cancer actions from January 2025 to July 2026, the three capabilities they supply, and the one they leave unfilled.

The commitment is real and it is funded, and it ran alongside proposed reductions: the FY2027 request of April 1, 2026 proposed roughly \$41.1 billion for NIH against \$48.7 billion in FY2026, though NCI would receive \$7.353 billion, an increase of \$8.9 million, and Congress added \$128 million for cancer research in the final FY2026 bill [18, 19, 20]. What none of it supplies is a sponsor-side method. Compute, data, agency review capacity, and faster site activation are all being built. The document a sponsor must file is still written the way it was written in 2015, and that is the gap Figure 1 states and this paper closes.

Table 5 and Figure 2 carry the same ten axes: the first as text a reader can cite, the second as a grid a reader can scan. The prior system is not described here as bad work. It is described as work whose unit is the team-month, and a team-month cannot produce an IND, two protocols, five bill versions and fourteen funding artifacts inside a single quarter. The remainder of this paper is the evidence for that sentence: §2 gives the method, §3 through §7 give the four deliverables and the review regime that cleared them, and §8 states what remains exposed.

d2-type / grid

Operating axis	Prior system	New system
Document production unit	Team-month	Prompt-hour
IND assembly time	Months across a group	Four days, one author
Peer review entry point	After completion	During development
Review latency	7 to 8 weeks best case	Same day, hour scale
Reviewers per round	Two to three humans	Three model manufacturers
Virtual trial cost	Above 120000 dollars	36330 dollars projected
Provenance record	Acknowledgement paragraph	Commit history plus DOI
Figure format	Raster, not reusable	Machine readable, reusable
Revision granularity	Whole manuscript	One file, one commit
Author count	Group, plus a CRO	One, plus the models

Row 10 is the axis the other nine follow from

Every cell carries a measured value from a deposited source, not an adjective. The prior column is white and the new column pale burgundy; the one emphasis cell is the axis that produces the other nine.

Figure 2. Ten operating axes on which the prior and the new trial system differ, each cell carrying a measured value from the deposited record.

Operating axis	Prior system	New system	Established in
Document production unit	Team-month	Prompt-hour	§2
IND assembly time	Months across a group	Four days, one author	§3
Peer review entry point	After completion	During development	§7
Review latency	7 to 8 weeks best case	Same day	§7
Reviewers per round	2 to 3 humans	3 model manufacturers	§7
Virtual trial cost	Above \$120,000 per run	\$36,330 projected	§6
Provenance record	Acknowledgement paragraph	Commit history plus DOI	§2
Figure format	Raster, not reusable	Machine readable, reusable	§2
Revision granularity	Whole manuscript	One file, one commit	§2
Author count	Group, plus a CRO	One, plus the models	Every deposit

Table 5. The ten operating axes of Figure 2 as citable text, with the section of this paper where each value is established.

The rest of the paper is organized as its evidence rather than as an argument. §2 gives the method and the two properties that make it repeatable. §3 through §6 give the four deliverables the method produced, each written to a similar length because each is load-bearing: the IND, the two trial protocols, the legislative framework, and the funding proposals. §7 gives the review regime that cleared all four and the measurements that justify trusting it. §8 states what remains exposed and what future work consists of, and §9 states what follows. A reader who wants only the numbers can read Tables 8 through 20 and stop.

1.1 What this paper is, and what it is not

This paper is a disclosure of a production method and a record of what that method produced. It is not a clinical result, a regulatory filing, or an enacted statute, and it makes no claim in any of those directions. The IND described in §3 is deposited, not filed. The protocols of §4 are written, not approved. The bill of §5 is drafted, not introduced. The applications of §6 are submitted, and none has yet been awarded.

What is claimed is narrower and checkable. First, that the four deliverables exist, are deposited under digital object identifiers, and were produced on the timescale the tables in this paper report. Second, that they were produced by one author directing a single model family under a single master prompt, and reviewed during development by two independent model manufacturers. Third, that the review regime of §7 is the only one available at this production rate, because the prior system's arithmetic does not permit it. Fourth, that the remaining obstacle to running the trial is institutional and financial rather than documentary. Each of the four can be falsified by a reader with the repository and no special access, and §3 names the four falsification routes for the dossier specifically.

The tone of the Federal record surveyed above is ambitious, and this paper is written to match it rather than to hedge against it. An administration that has pointed executive orders, appropriations and agency capability at the intersection of artificial intelligence and cancer within eighteen months is entitled to expect sponsors to move at a comparable rate, and the purpose of this disclosure is to show that a sponsor can.

2 Methods

The method is one master prompt, decomposed by the model into sub-prompts that the model then executes in order, with every generated file committed and pushed before the next is written. Earlier 2026 papers in this repository reached a similar effect through a sequence of user prompts, one per stage: the regulatory adaptations of March 2026, the national platform paper, and the site documentation package were all built that way, and the author had to be present at each stage boundary to issue the next instruction [21]. Under the master prompt schedule the author is present once, at the start, and thereafter watches the branch. The schedule used here has eight stages: five that produce machine-readable figure specifications, one per diagram platform, and three that produce the paper itself at increasing quality, draft, full and final. Figure 3 draws that schedule as a single activity and marks where the concurrency actually is.

Two properties of the schedule matter more than its length. The first is that the twenty-five figure slots are fixed before the fork, so five specification stages can run against one plan without any of them invalidating another; the second is that the unit of work is one file rather than one stage, so the branch carries a usable state after every write rather than after every stage. Figure 4 draws one turn at that granularity. Table 6 counts how many separate deposited projects have used each element of the method, which is the basis on which it is offered as reliable rather than as novel.

Method element	Projects	First use	Evidence
Master prompt, single turn	3	Jul 2026	funding/pdac-funding-applications/prompts
AI-generated sub-prompt schedule	3	Jul 2026	funding/capitalization-plan/sub-prompts
Five-platform specification set	3	Jul 2026	Five directories per project
Draft, full, final staging	8	Mar 2026	Every paper directory since March 2026
One file, one commit, pushed live	3	Jul 2026	The branch history of each build
Machine-readable figure reuse	3	Jul 2026	Specifications re-read, not re-derived
AI peer review round	5	Nov 2025	§7, Table 24
DOI deposit per artifact	30 or more	2024	new-trial-system/abstracts/README.md

Table 6. How many separate deposited projects have used each element of the method, and where the evidence for each of those counts is held.

The second half of the method is storage. Every figure in this paper exists first as a specification: a perspective statement, a two-line caption, valid source in the platform's own syntax, a construction table of absolute coordinates, an edge-routing paragraph, and a value table naming the repository file each number came from. That format costs about 6 to 9 kilobytes per figure and can be read back as the input to the next paper; a raster of the same figure costs 180 to 900 kilobytes and can be read back as nothing. Figure 5 puts the two records side by side, and Table 7 states what each stored field buys.

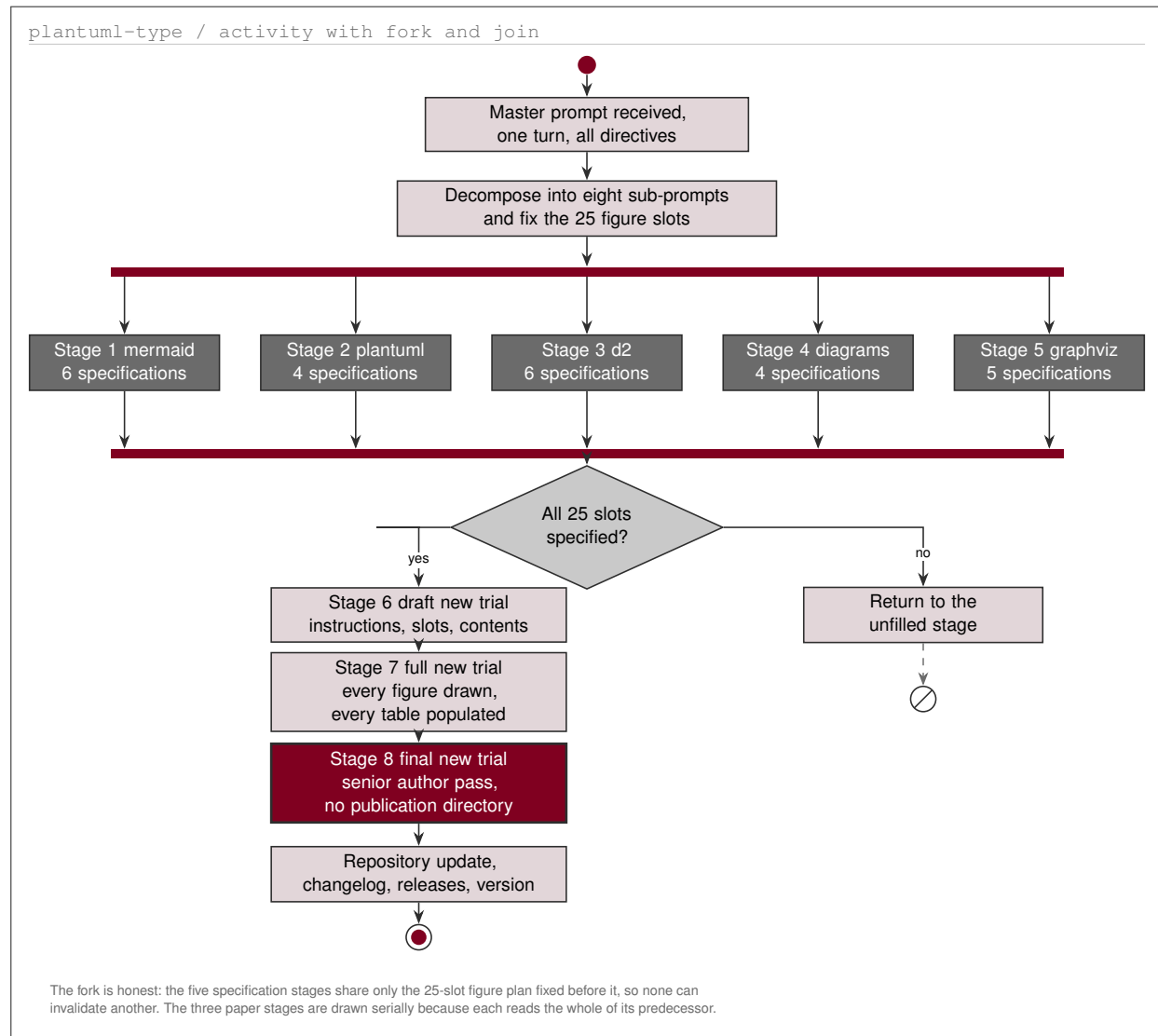


Figure 3. The eight-stage build as one activity: five diagram stages fork and join on a shared specification, then three paper stages run strictly in order.

Property	Specification	Raster	Consequence for the next paper
Size on disk	6 to 9 kilobytes	180 to 900 kilobytes	The whole figure set fits in one raster's space
Caption	Two lines, exact	None	Captions are diffable across versions
Coordinates	Absolute, stated	None	A node can be added without a re-layout
Routing	Stated per edge	None	Overlap claims are auditable
Values	Sourced per cell	None	No number in a figure is unattributable
Reuse	Read back as input	Re-drawn by hand	The next paper starts from this one

Table 7. What a machine-readable figure specification stores that a raster does not, and what each stored field buys the next paper.

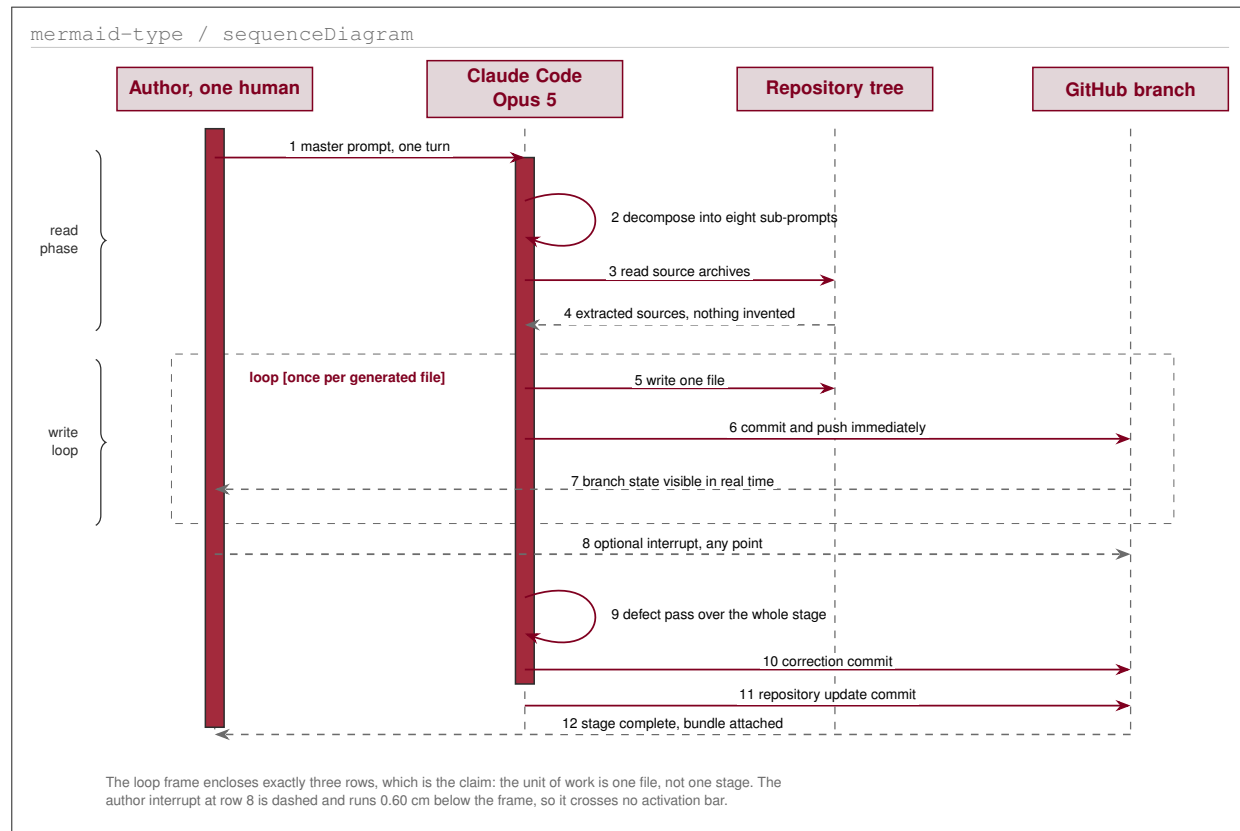


Figure 4. One generation turn from master prompt to pushed commit, with the four participants, the two return paths, and the point the author can stop it.

Two properties of this method are worth separating from its mechanics, because they are what make it repeatable by somebody else. The first is that nothing in the schedule requires the author to be a good writer at speed. The specification stages produce structured objects, not prose, and the paper stages read those objects; the quality of the figures is therefore a property of the specifications rather than of the drafting session that used them. The second is that the branch is the record. There is no separate build log, no note-taking file, and no reconstruction after the fact: the order in which the twenty-five specifications, the three stage bundles and the eleven section files appear on the branch is the order in which they were written, and it is checkable by anybody with the repository [22].

2.1 What the method costs to adopt

The method is offered as adoptable, so its cost should be stated. It requires one person, a repository, a model subscription, and a LaTeX toolchain. It does not require a team, a project manager, a document management system, or a version control convention beyond one file per commit. The eight-stage schedule is a text file; the figure specifications are text files; the paper is a text file set. Nothing in the method is proprietary to the author, and the two directories that record it, `new-trial-system/prompts` and `new-trial-system/sub-prompts`, are written to be copied rather than admired.

Three failure modes are worth naming for anybody adopting it. The first is fixing the figure plan too late: if the twenty-five slots are not settled before the specification stages fork, the stages contradict one another and the paper rennumbers, which is the layout-instability defect the specifications are written against. The second is letting a stage hold work back for a batch commit, which destroys the property that makes the branch a record. The third is writing prose before the specifications exist, which produces figures that illustrate paragraphs rather than paragraphs that read figures; every figure in this paper was specified before the

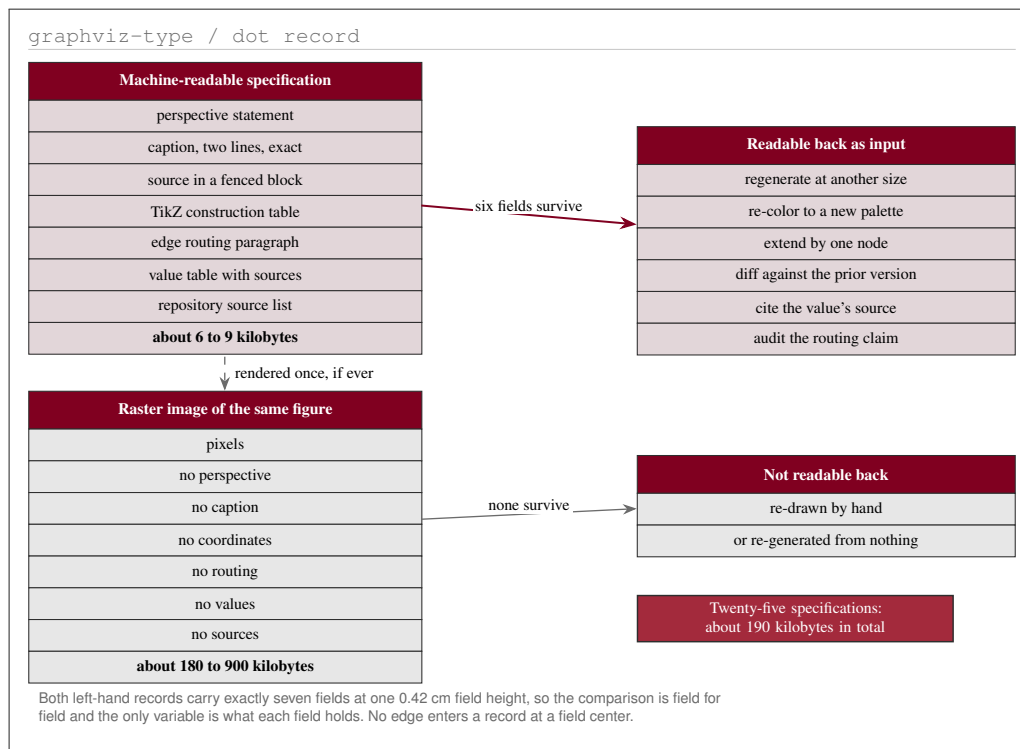


Figure 5. Two records for one figure: the machine-readable specification and the raster image, with the six fields only the first returns as input.

section that discusses it was written, and the difference is visible in how often the prose says what a figure shows rather than restating it.

2.2 Provenance as an output, not a byproduct

The word provenance is used deliberately throughout this paper. In the prior system, provenance is an acknowledgement paragraph: a claim about who did what, made after the fact, unverifiable in principle. In the new system it is the commit history, which is produced as a side effect of the production rule rather than written afterward, and which records the order, the timing and the content of every write. That distinction is what makes the timing claims of §3 through §6 checkable rather than assertable, and it is the property a funder should test first, because it is the cheapest to test and the one everything else depends on.

3 IND

3.1 What was filed, and under what authority

The Investigational New Drug application deposited July 1, 2026 supports a Phase 1, first-in-human, combined IND and Investigational Device Exemption investigation of on-premises large language model directed eight-arm robotic pancreaticoduodenectomy with perioperative daraxonrasib (RMC-6236) in patients with KRAS G12-mutated pancreatic ductal adenocarcinoma [23]. The dossier states its own authority in its opening subsection: the investigation “is filed under 21 CFR §312 for the investigational drug and is coupled to a significant-risk early-feasibility device study under 21 CFR §812 for the LLM-directed robotic platform; the two pathways are coordinated within a single integrated protocol because the device conducts the resection that generates the operative and pathology inputs to the daraxonrasib restart advisory” [23]. That sentence is doing regulatory work rather than rhetorical work: it is the reason a single dossier can carry both a drug and a device without either pathway being subordinate to the other.

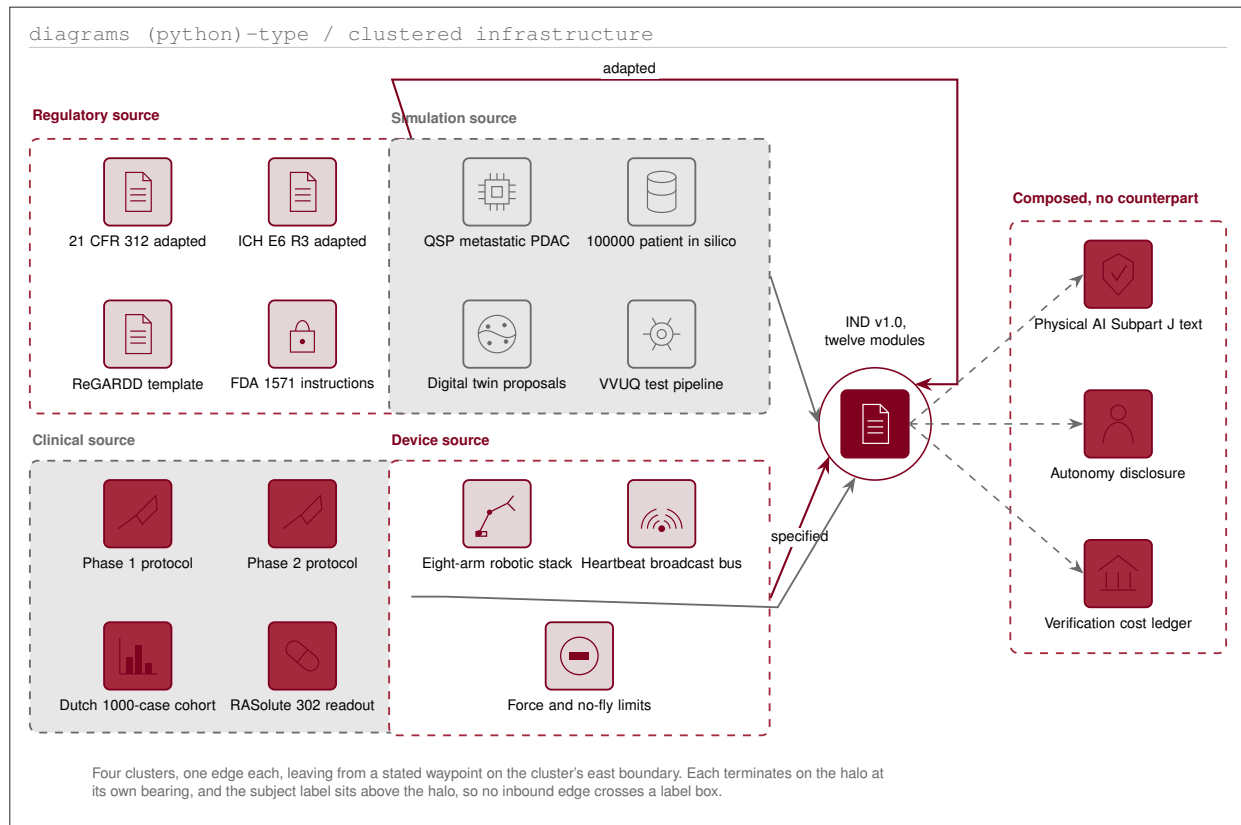


Figure 6. Four source clusters feeding twelve IND modules, and the three modules that had no prior-system counterpart and were composed instead.

The clinical problem the dossier addresses is stated with the same discipline. PDAC is the third leading cause of cancer death in the United States with total five-year overall survival below 13 percent [24], and the pancreaticoduodenectomy requires resection and reconstruction of four luminal structures and dissection around five named vessels. The 2025 Dutch nationwide cohort of 1000 robotic pancreaticoduodenectomies, which anchors the contemporary human baseline, reported a conversion rate of 10.1 percent, Clavien-Dindo grade III or higher complications of 41.3 percent, International Study Group on Pancreatic Surgery grade B or C fistula of 24.4 percent, and 30-day mortality of 3.9 percent [25, 26]. Figure 6 shows what the dossier was assembled from, and Table 8 lists the twelve modules it was assembled into.

Module	Position	Characters	What it establishes
Cover letter	1	19,070	Transmittal and sponsor identity
Table of contents	2	generated	ReGARDD content order
FDA forms 1571 and 1572	3	32,067	The codified form set
Introduction	4	82,102	§312.23(a)(3)(i) introductory statement
General investigational plan	5	76,989	§312.23(a)(3)(iv) plan
Investigator's brochure	6	47,368	§312.23(a)(5) brochure
Proposed clinical research	7	76,193	§312.23(a)(6) protocols
Chemistry, manufacturing, control	8	56,370	§312.23(a)(7) CMC
Pharmacology and toxicology	9	55,719	§312.23(a)(8) nonclinical
Previous human experience	10	48,045	§312.23(a)(9) prior exposure
Additional information	11	47,087	§312.23(a)(10)
Relevant information and back matter	12	53,647	§312.23(a)(11) and references

Table 8. The twelve modules of the deposited IND in ReGARDD content order, their character count at deposit, and the codified requirement each satisfies.

3.2 The clock

The dossier was built in four stages over four days: a figure catalog, a draft, a full version, and a senior-author final pass, deposited July 1, 2026 at repository version v4.3.0 [23]. The prior system's published calendar for the same twelve modules is measured in months, and its stages carry different names: regulatory drafting, nonclinical write-up, chemistry compilation, internal quality control and legal review, then sponsor sign-off. Figure 7 places both regimes on one axis with a single break glyph where the two scales part, and Table 9 gives the day each module closed against the prior system's name for the same work.

Module	Repository source read	Window	Prior-system counterpart
Introduction and plan	Adapted 21 CFR 312, ICH E6(R3)	Day 1	Regulatory drafting, months
Investigator's brochure	Phase 1 protocol, RASolute 302	Day 2	Nonclinical write-up
Proposed clinical research	Phase 1 and Phase 2 protocols	Day 2	Regulatory drafting
Chemistry, manufacturing, control	Public daraxonrasib record	Day 2	CMC compilation
Pharmacology and toxicology	QSP and digital twin record	Day 3	Nonclinical write-up
Previous human experience	RASolute 302, Dutch cohort	Day 3	Internal quality control
Figures, tables, back matter	The 22-figure catalog	Day 4	Sponsor sign-off

Table 9. Each IND module, the repository source the generation read, the day it closed, and the prior system's name for the same work.

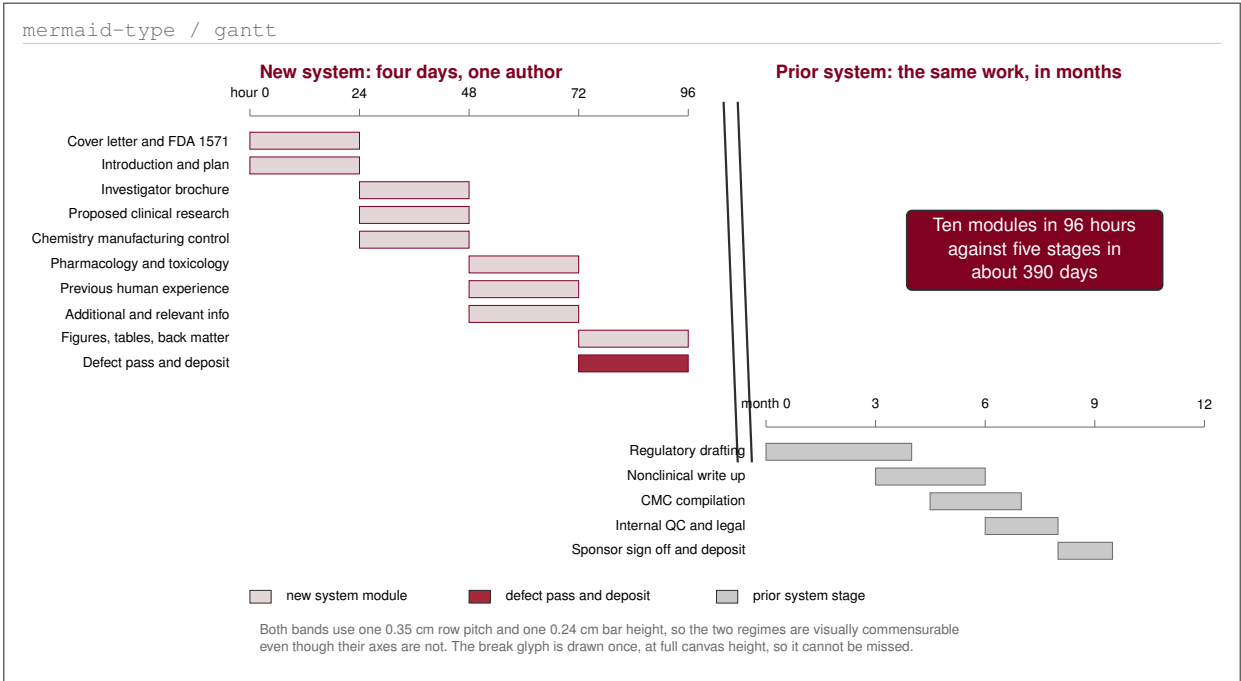


Figure 7. Twelve IND modules on one clock: the new system in hours across four days, and the prior system’s published calendar for the same work, in months.

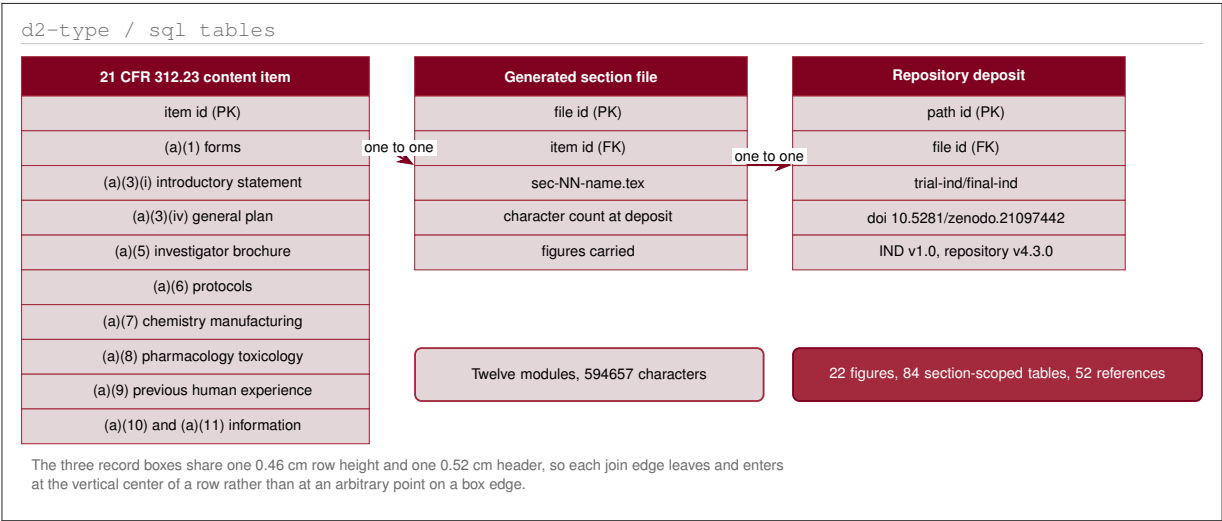


Figure 8. Twelve codified IND content items joined to the section file that satisfies each, and to the repository path where that file is deposited.

3.3 Complexity and volume

The complexity feat is not the speed on its own. It is that the speed was reached while the dossier stayed internally consistent across 594,657 characters of regulatory prose, 52 distinct 21 CFR cross-references, 22 figures in a single document sequence, 84 section-scoped tables, and 52 bibliography entries, with every figure drawn in vector form and no raster anywhere. A reviewer opening the IND at §312.23(a)(8) and following its cross-reference into the general investigational plan finds the same dose levels, the same stopping rules and the same device limits stated the same way, because all twelve modules were written against one another rather than by separate hands. Figure 8 is the crosswalk that makes that checkable, and Table 10 states the volume.

Volume measure	Deposited IND	What it means for a reviewer
Section files	12	One per ReGARDD content position
Total characters across sections	594,657	The dossier's written weight
Figures, single document sequence	22	Every figure numbered once, end to end
Tables, section-scoped numbering	84	Table 3.1, Table 5.2, and so on
Distinct 21 CFR cross-references	52	Codified coverage that can be spot-checked
Bibliography entries	52	Every claim carries a source
Raster images	0	Every figure is vector and re-renderable
Authors	1	With model assistance, disclosed
Elapsed production time	4 days	Against months in the prior system

Table 10. The volume one author reached in four days, measured on nine axes, and what each measure means to a regulatory reviewer.

Figure 9 states the sponsor's residual risk as logic rather than as a list. The two clinical branches are AND gates, so a hold on clinical grounds requires both a signal and a failure to respond to it, and the protocol's pause rule is written to break that conjunction. The documentation branch is an OR gate, where any one of four omissions suffices: insufficient chemistry information, a brochure that omits a known risk, consent that lacks the autonomy disclosure, or a sponsor report filed outside its window. That branch is exactly the class of failure a system which can regenerate a complete, internally consistent dossier in four days is built to eliminate, because the cost of a full re-issue after a finding is four days rather than a quarter [23, 27].

3.4 What the volume did not cost

Volume of this kind invites one obvious objection: that a dossier written at this rate must be padded, and that 594,657 characters is a measure of verbosity rather than of coverage. The objection is answerable, and the answer is in the structure rather than in the prose. Every one of the twelve modules is written to a codified content position, so a module cannot grow by adding material that belongs somewhere else; the 52 distinct 21 CFR cross-references are checkable one by one against the regulation; the 84 tables are section-scoped rather than document-scoped, so a reviewer reading §312.23(a)(7) finds Table 6.1 and not Table 43; and the 22 figures are numbered in a single document sequence, so a cross-reference from the pharmacology module to a figure in the plan resolves without ambiguity [23].

The second objection is that speed of this kind must come from copying. It partly does, and the copying is disclosed: the proposed clinical research module carries the Phase 1 protocol, the general investigational plan carries the Phase 2 forward path, and the adapted 21 CFR and ICH text supplies the regulatory scaffolding, all of which are separately deposited and separately dated [28, 29, 21]. What was not copied is the part Figure 6 marks as composed: the Physical AI Subpart J text, the autonomy disclosure, and the verification cost ledger. None of the three has a prior-system counterpart to copy from, because no prior IND has been filed for a device that generates the code it executes.

The third objection is the serious one: that a dossier assembled this fast cannot have been checked. It was checked, twice, by two model manufacturers independent of the one that produced it, at defined milestones and before deposit, which is the review regime §7 describes and measures [30, 31]. That regime is not equivalent to a regulatory review and does not claim to be. It is equivalent to the internal quality control stage the prior system runs before sponsor sign-off, and it runs in hours rather than in the sixty days that stage is typically budgeted.

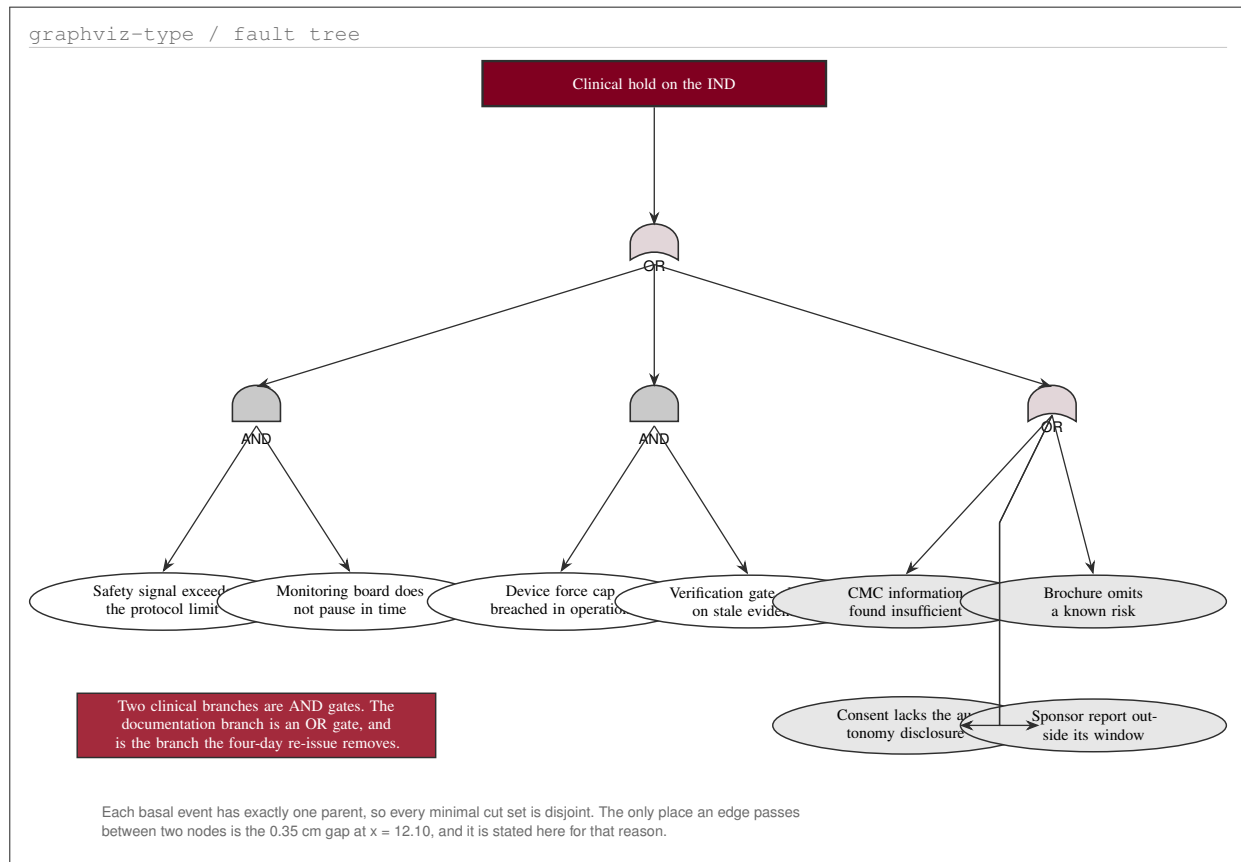


Figure 9. The clinical hold as a top event over eight basal failures, with the two AND gates that make a hold require a coincidence rather than a fault.

3.5 What a reviewer can check in an afternoon

The practical test of a dossier is not its length but how quickly a reviewer can falsify it. Four checks are available to anyone with the deposited files and no special access. First, open Figure 8's crosswalk and pick any codified content item; the named section file either satisfies it or it does not. Second, take any of the 52 CFR cross-references and read the cited paragraph; the dossier either states the requirement correctly or it does not. Third, take any number that appears in more than one module, the day-30 safety window, the 3 N per-arm force cap, the $n = 18$ ceiling, and check that every occurrence agrees. Fourth, read the four-day production claim against the commit history of trial-ind, which records when each module was written and pushed. Each of the four is a falsification route, and each is open because the artifacts are deposited rather than described.

3.6 The prior system's own account of the same work

It is worth being precise about what the prior system does with these twelve modules, because the comparison in Figure 7 is otherwise an assertion. In the prior system a Phase 1 IND for a combined drug and device investigation is produced by a regulatory affairs function working from a protocol supplied by clinical development, a chemistry section supplied by manufacturing, a nonclinical section supplied by toxicology, and a brochure assembled from all three. Each of those inputs arrives on its own schedule; the regulatory function reconciles them; legal and quality review the reconciliation; and the sponsor signs. The five stages of Figure 7's lower band are that sequence, and the months on its axis are the intervals between handoffs rather than the time spent writing.

That is the structural reason the new system's four days is not merely a faster version of the same thing. There are no handoffs. The general investigational plan and the chemistry module were written against one another within the same day rather than reconciled across two functions in the same quarter, which is why a number that appears in both appears identically in both without a reconciliation step. The comprehensive cross-reference discipline reported in Table 10, 52 codified references across twelve modules, is a consequence of that adjacency rather than an achievement on top of it.

The corresponding loss is also structural and should be stated. A single author cannot supply the independent judgment that four functions supply, and no amount of internal consistency substitutes for it. That is precisely the gap §7 fills with two independent model manufacturers reviewing during development, and it is the reason the review regime is reported as part of the method rather than as an afterthought.

3.7 What the dossier would need to become a filing

Nothing in §3 is filed. Converting the deposited dossier into an active IND requires four things the sponsor does not hold: a sponsor-investigator of record willing to sign Form FDA 1571 for the investigation as described, an institutional review board of record willing to review a combined IND and IDE submission carrying a Physical AI consent opt-out, a drug supply agreement for daraxonrasib, and a device acceptance certificate for the eight-arm platform described in §4. Each is an institutional instrument rather than a document, and each is named in the register of §8.

The dossier is written so that acquiring those four does not require rewriting it. Every module is separately compilable, every figure is drawn from a specification rather than from a rendering, and every table is section-scoped rather than document-scoped, so a change forced by a site's particulars touches one module and rennumbers nothing. That property is worth more to a sponsor than the four-day production figure, because a filing is amended far more often than it is written. That property also sets what a review round costs. A dossier assembled in four days is answerable to §7's clock rather than to a journal's: its twelve modules were read by three model manufacturers against the codified content items before the deposit, not seven to eight weeks after it, so every correction landed in the filed document rather than in an erratum. For a disease with roughly 142 US deaths a day [24], the difference between correcting a dossier before it is filed and correcting it after is a quarter of a year of enrollment that either happens or does not.

4 Trial Protocol

4.1 The Phase 1 study

The Phase 1 protocol, version 1.0.0, deposited June 21, 2026, describes itself as an “open-label, single-arm, dose-finding study using a standard 3+3 escalation across three daraxonrasib dose levels (DL1 through DL3) with staggered (sentinel) enrollment”, planned for up to $n = 18$ participants [28]. A mandatory gate precedes any patient-facing robotic activity: “a mandatory Phase 0 simulation-validation gate (≥ 1000 simulated procedures across ≥ 2 independent frameworks, with a Unified Safety Level (USL) rating ≥ 7.0) precedes any patient-facing robotic activity, and prespecified pause and stopping rules govern progression” [28]. The intervention is stated with the same numeric precision: perioperative daraxonrasib with a planned preoperative pause and an LLM-advised postoperative restart at T+7, T+14, or T+21 days keyed to fistula status and drug trough, combined with an eight-arm robotic Whipple performed under continuous human oversight as a Class II collaborative device, with a 10 kHz heartbeat broadcast bus, a cross-arm emergency-stop budget at or below 3 ms, per-arm tip-force caps at or below 3 N, a cumulative cross-arm cap at or below 18 N, and vascular no-fly gating around the superior mesenteric and portal veins [28]. Table 11 is the deposited synopsis in full.

Element	Phase 1, first-in-human, v1.0.0, June 21, 2026
Design	Open-label, single-arm, 3+3 escalation, DL1 to DL3, staggered sentinel enrollment
Sample size	Up to $n = 18$
Primary endpoints	Dose-limiting toxicity rate during escalation; device- and procedure-related serious adverse event rate through day 30, including Clavien-Dindo grade III or higher
Key secondary	R0 rate at day-90 pathology; ISGPS grade B or C fistula rate; 90-day mortality; progression-free survival; overall survival to 24 months
Population	Adults 18 or older, resectable or borderline-resectable KRAS G12-mutated PDAC, ECOG 0 or 1, adequate organ function
Gate	Phase 0: at least 1000 simulations across at least 2 frameworks, Unified Safety Level at or above 7.0
Device limits	10 kHz heartbeat bus; cross-arm stop at or below 3 ms; per-arm tip force at or below 3 N; cross-arm at or below 18 N
Duration	Up to 28 days screening, surgery on day 0, day 1 to 7 acute window, day 30 primary safety window, day 90 pathology, follow-up every 12 weeks to 24 months
Regulatory basis	IND 21 CFR §312 plus IDE 21 CFR §812, Physical AI Subpart J

Table 11. The Phase 1 protocol synopsis as deposited, with the gate that must close before any patient-facing robotic activity begins.

Figure 10 takes the participant’s point of view rather than the study’s. Nine forward states run down a left column at a constant 1.30 cm pitch, two exception states sit in a right column 8.20 cm away, and each of the fourteen transitions carries a guard written as a quantity that can be evaluated at the bedside. The guards leaving any one state partition the space: the complement of “KRAS G12 present and ECOG 0 or 1” is the screen-failure guard, the complement of “resection complete, no intraoperative stop” is the force-cap guard, and the complement of “fistula grade below B and drug trough in range” is the dose-limiting toxicity guard. No participant can therefore occupy two successors at once, and no state can deadlock.

4.2 The Phase 2 study

The Phase 2 protocol, version 1.1.0, deposited June 23, 2026, two days after its predicate, is a “multicenter, randomized 1:1, parallel-group, controlled, open-label study with BICR of imaging, central masked pathology, and blinded endpoint adjudication, conducted at eight high-volume HPB centers with a planned

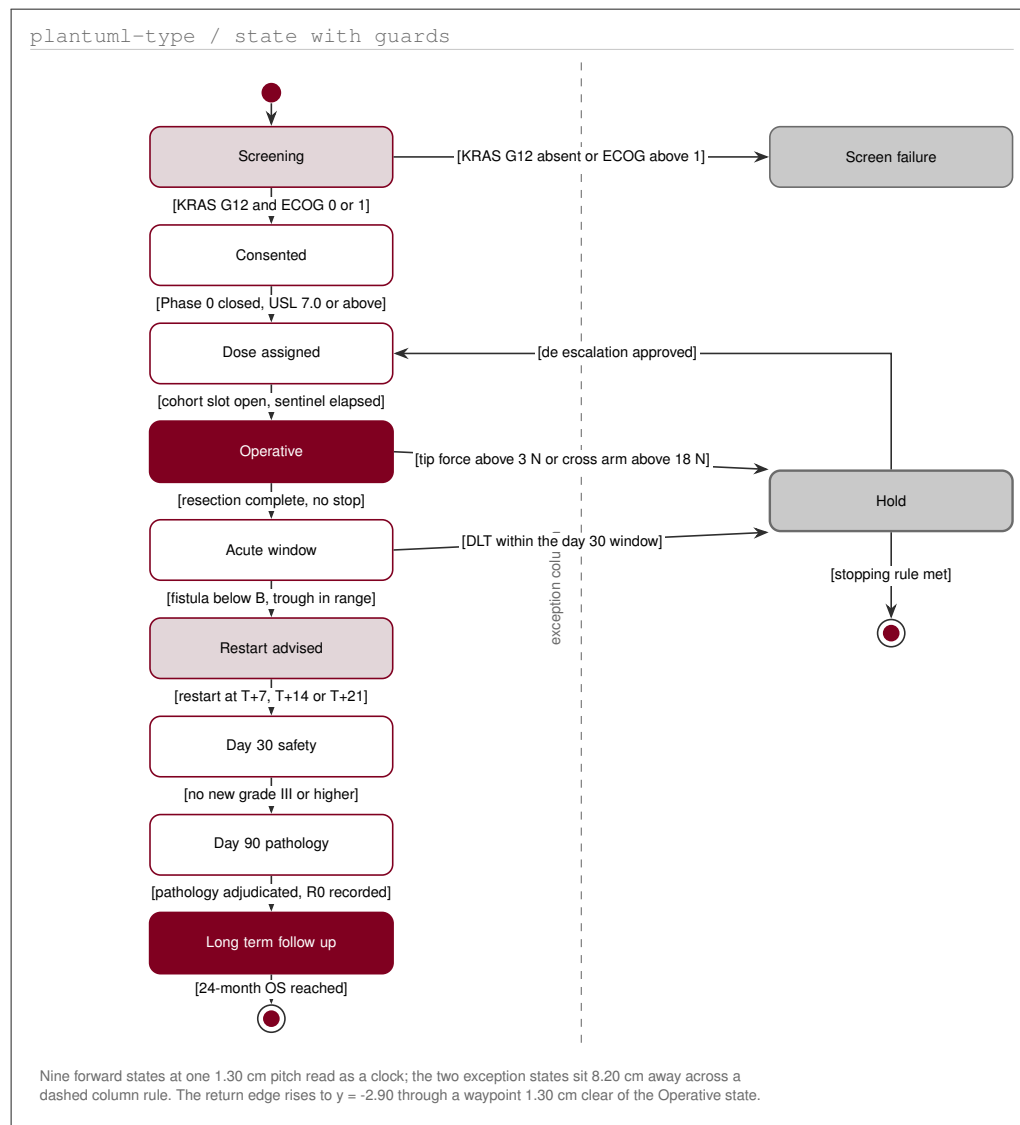


Figure 10. One participant from screening to the 24-month endpoint, with the guard on every transition and the two conditions that route to a study hold.

sample size of $n = 220$ randomized participants” [29]. It states its own dependence on the predicate plainly: the first-in-human study “established the daraxonrasib RP2D and demonstrated the feasibility and hard cyber-physical safety of an on-premises LLM-directed robotic Whipple, but it was not powered for efficacy and explicitly deferred the comparative question to a randomized trial” [29]. The trial targets approximately 140 progression-free survival events for 85 percent power at a two-sided alpha of 0.05 to detect a hazard ratio of 0.60, median PFS 14.0 months control against 23.3 months experimental, with a single interim analysis at approximately 60 percent of events using O’Brien-Fleming spending. Table 12 is the Phase 2 synopsis, and every randomized element in it depends on a Phase 1 result.

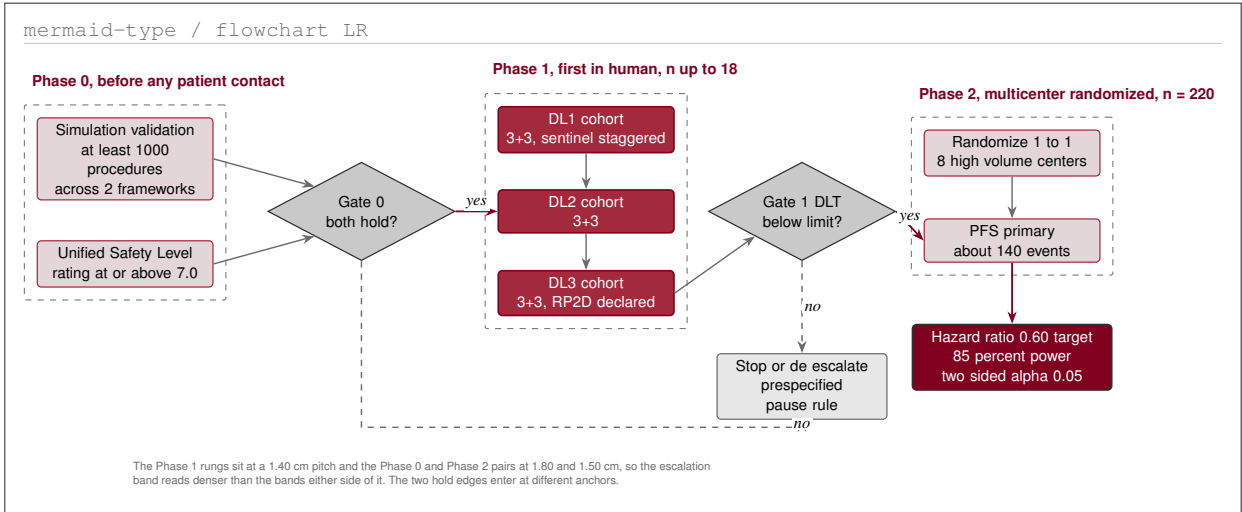


Figure 11. From the Phase 0 simulation gate through three 3+3 dose levels to Phase 2 randomization, with the quantity that must hold on each rung shown.

Element	Phase 2, randomized, v1.1.0, June 23, 2026
Design	Multicenter, randomized 1:1, parallel-group, controlled, open-label, with blinded independent central review
Sample size	n = 220, approximately 110 per arm, eight high-volume HPB centers
Primary endpoint	Progression-free survival per RECIST 1.1, investigator assessed, BICR as sensitivity analysis
Key secondary	Overall survival; R0 rate; ISGPS grade B or C fistula rate; major pathologic response; KRAS ctDNA clearance at week 12
Statistics	About 140 PFS events, 85 percent power, two-sided alpha 0.05, target hazard ratio 0.60, median 14.0 against 23.3 months
Interim analysis	One, at about 60 percent of events, O’Brien-Fleming spending, independent monitoring board
Stratification	Resectability, KRAS allele G12D against other G12, neoadjuvant therapy, site
Predicate	Phase 1 v1.0.0, which established the RP2D and the safety architecture

Table 12. The Phase 2 protocol synopsis as deposited, and the Phase 1 predicate whose result each randomized element of it depends upon.

Figure 11 draws the clinical progression that joins the two documents, and Figure 12 draws the documentary one. Four of the twelve Phase 1 sections were carried into Phase 2 unchanged, four were replaced outright because the study is no longer single-arm, and four elements exist in the Phase 2 document only because randomization requires them. Table 13 states the elapsed production time for both.

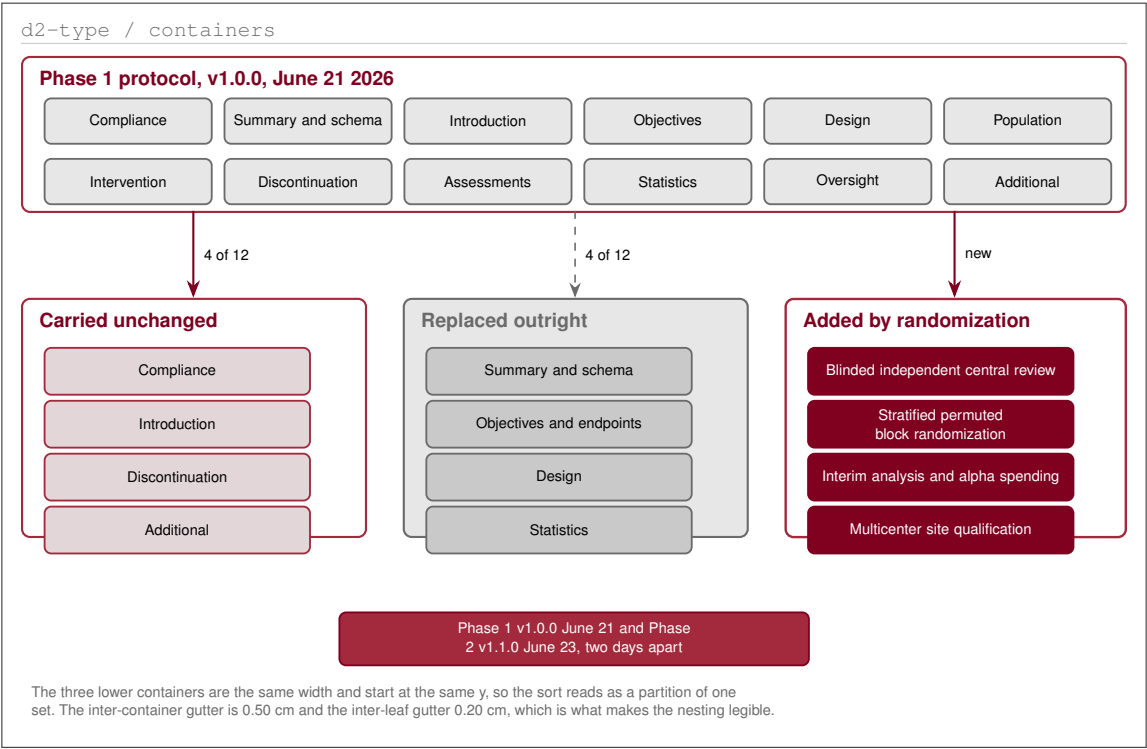


Figure 12. The twelve Phase 1 protocol sections sorted into what Phase 2 carried unchanged, what it replaced, and what randomization made necessary.

Document	Version	Deposit	Characters	Figures	Elapsed against the prior system
Phase 1 protocol	v1.0.0	Jun 21, 2026	169,878	20	2 days against 6 to 12 months
Phase 2 protocol	v1.1.0	Jun 23, 2026	199,421	22	2 days against 6 to 12 months
Both, combined	Two	Jun 21 to 23	369,299	42	4 days against a full design cycle

Table 13. Both protocol deposits, their versions, character counts and figure counts, and the elapsed production time against the prior system.

4.3 The site the protocol assumes

Figure 13 draws the equipment a participant is operated within, and the one boundary that makes the protocol’s on-premises claim checkable. Four layers sit inside the building: oversight, control, device, and record. One node sits outside it. The prohibited path from the on-premises model host to any external network is the only struck-through edge in this paper, and a site that cannot demonstrate that edge struck is not running this protocol, whatever else it satisfies.

The boundary is not a preference. It is what makes the consent language operable: a participant told that an on-premises model advises the operation can verify the claim by asking what crosses the building line, and a monitor can audit it by inspecting one network configuration rather than a vendor’s assurances [28, 32]. The same boundary is what the autonomy disclosure duty in §5 attaches to.

4.4 Why two protocols rather than one

A Phase 1 protocol and a Phase 2 protocol two days apart is an unusual pairing, and the reason is structural rather than opportunistic. A first-in-human study of a device that advises an operation has to answer a

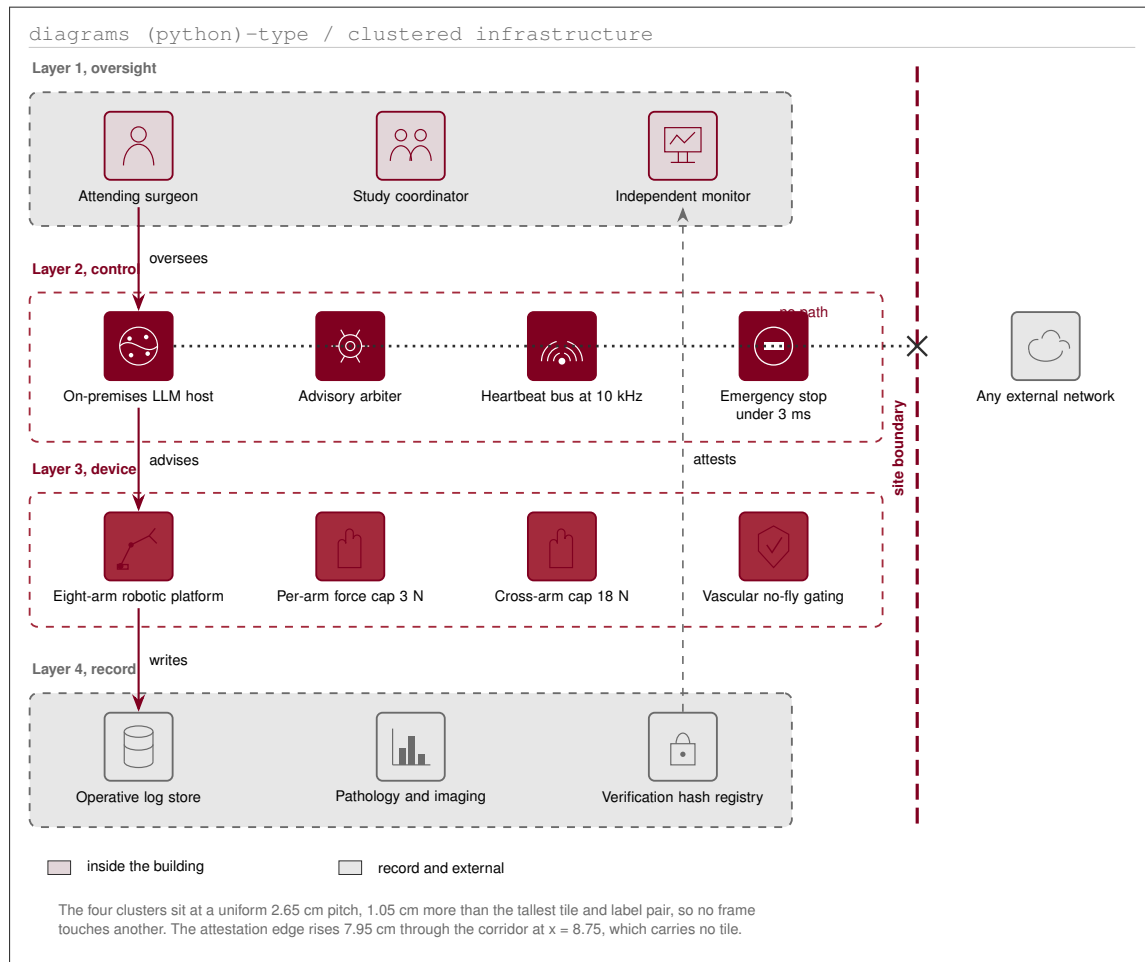


Figure 13. The four-layer on-premises site stack, and the single boundary across which no inference request, model weight, or patient record passes.

question the drug arm alone cannot answer: whether the advisory changes the operation in a way that a randomized comparison would later detect. Writing the randomized study first, as a target, forces the first-in-human study to collect what the randomized study will need. The Phase 2 document therefore fixes the primary endpoint, progression-free survival per RECIST 1.1 with blinded independent central review as a sensitivity analysis, and the Phase 1 document collects the operative and pathology inputs that endpoint depends on [28, 29].

The pairing also exposes what the prior system's sequencing hides. In the prior system the Phase 2 protocol is written after the Phase 1 result is known, which means the Phase 1 case report form was designed without knowing which of its fields would matter. Here both documents were written against one another, and Figure 12 shows the consequence: only four of the twelve Phase 1 sections had to be replaced when the study randomized, and none of the four is a data-collection section. Compliance, introduction, discontinuation and additional considerations carried unchanged. Summary, objectives, design and statistics were replaced, which is exactly the set a change from single-arm dose-finding to randomized comparison should replace and no more.

4.5 What the two protocols assume about the site

Both protocols assume a site that does not yet exist in the form they require, and both say so. The Phase 1 protocol assumes one qualified high-volume hepato-pancreato-biliary center with an eight-arm robotic platform, an on-premises model host inside the building boundary of Figure 13, a pathology service able to

adjudicate margin status at day 90, and an institutional review board willing to review a combined IND and IDE submission with a Physical AI consent opt-out [28]. The Phase 2 protocol assumes eight such centers, plus central masked pathology, blinded endpoint adjudication, and a central randomization service with four stratification factors [29].

None of those instruments can be obtained by a sponsor of 2.6 full-time equivalents acting alone, which is the point §6 and §8 return to. The protocols are written to be executable by an institution that has them, not to be executable by their author. That is a deliberate limit on what document production can accomplish, and stating it here is more useful than implying otherwise: the documents remove the drafting bottleneck and they do not remove the institutional one.

4.6 The safety architecture in one paragraph

A reader who takes only one thing from §4 should take the safety architecture, because it is the part a clinician will test first. Four independent limits act on the operation at once. A heartbeat broadcast at 10 kHz means the control layer knows within 100 microseconds that an arm has stopped reporting. A cross-arm emergency stop budget at or below 3 ms means that when any arm asserts a stop, every other arm has halted before a human could have noticed the condition. Per-arm tip-force caps at or below 3 N and a cumulative cross-arm cap at or below 18 N mean the platform cannot apply more force than the caps allow even if every controller above them is wrong. Vascular no-fly gating around the superior mesenteric and portal veins means the geometric region in which those forces would be most dangerous is excluded before any of the other three limits is tested [28].

Those four are cyber-physical rather than advisory: none depends on the model being correct, and none depends on the surgeon reacting in time. The model's role sits above them, in the perioperative restart advisory and the operative guidance, and Figure 10's guard on the transition out of the operative state is written against the force caps rather than against the model's confidence. That ordering is the protocol's answer to the trust question §5 raises: the clinician is asked to rely on the model for advice and on the hardware for safety, and the two reliances are separable by design.

4.7 What the two protocols would need to open

The instruments the protocols require are the same class as those the IND requires, and the register in §8 lists all seven together. For the Phase 1 study the binding ones are a qualified high-volume hepato-pancreato-biliary center, an operating room with the eight-arm platform installed and accepted, a pathology service able to adjudicate margin status at day 90, and an institutional review board of record. For the Phase 2 study the binding ones are eight such centers plus central masked pathology, blinded endpoint adjudication, and a central randomization service.

The protocols are deliberately not written down to what one sponsor could run alone. A single-site, unblinded, unadjudicated study would have been easier to open and would not have answered the comparative question, and the Phase 1 document says as much when it defers that question to a randomized trial [28, 29]. Writing to the study that answers the question and then stating plainly what is missing is more useful to a funder than writing to the study that could be opened tomorrow. Both protocols carry the review provenance of the dossier of §3. The two documents and their shared statistical plan were frozen, hashed and read by three independent model reviewers, so the guards of Figure 10 and the rungs of Figure 11 were checked against each other before either was deposited. A protocol is where a late correction costs most, because a changed eligibility criterion or dose-limiting toxicity window invalidates the enrollment already done under it, and §7 prices that delay.

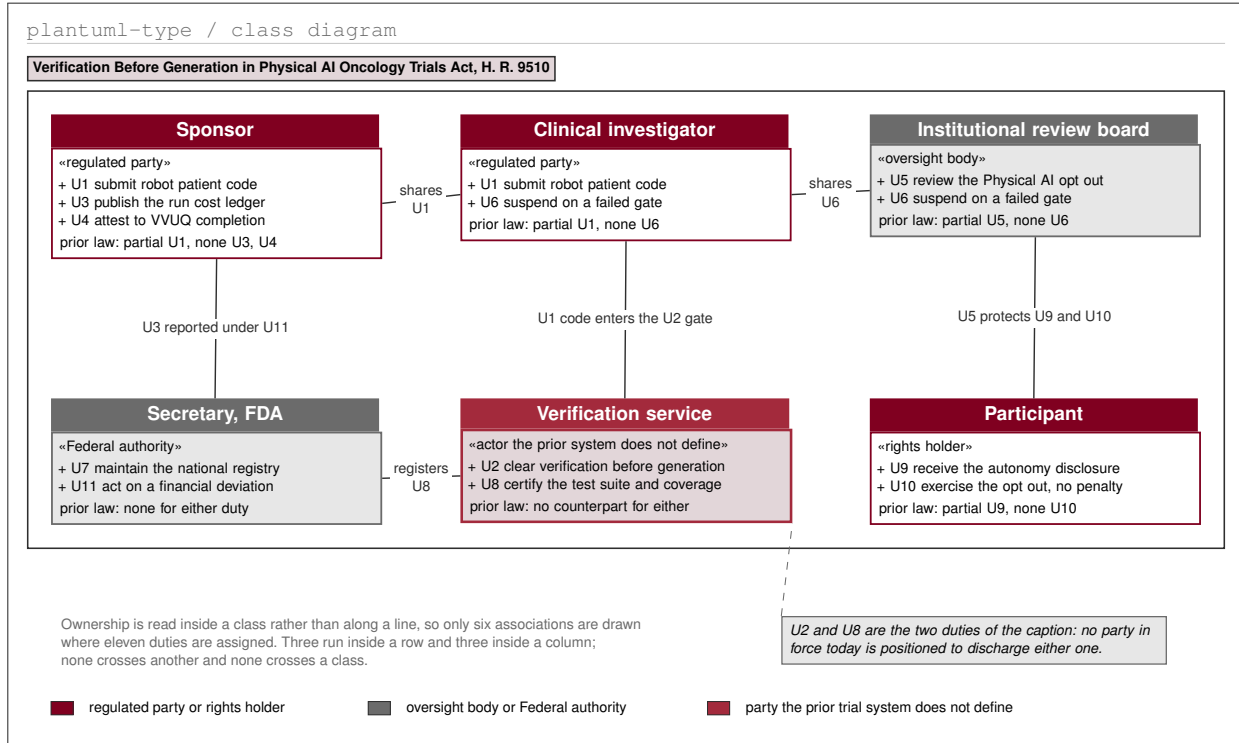


Figure 14. Six actors and eleven statutory duties, with the two duties that no actor recognized by the prior trial system is positioned to discharge.

5 Legislation

5.1 Who owes what

Five versions of one bill and four companion documents were deposited between May 30 and June 17, 2026. The bill is H. R. 9510, the Verification Before Generation in Physical AI Oncology Trials Act of 2026, and by version 5.0 it is a financial data amendment to the Federal Food, Drug, and Cosmetic Act [33]. Its findings state the operative requirement and the reason for it in one sentence each. The requirement: Congress finds “that an automated process must verify robot-patient interaction code before that code is generated or executed in a Physical AI oncology clinical investigation” [33]. The reason: “In a disembodied pipeline an error yields a wrong number a human can catch downstream; in an embodied surgical system the same error class yields a wrong motion a human cannot catch in time. No Federal law requires the verification of robot-patient interaction code to clear before that code is generated or executed, because the order of operations is not regulated” [33, 34].

The companion bill supplies the vocabulary the requirement needs. It defines verification as “confirmation that code was built correctly: implementation conforms to its specification” and validation as “confirmation that the right code was built, that is, the result agrees with an independent reference of reality”, both against ASME V&V 40, and it defines the class the gate applies to: an artificial intelligence system “whose output materially influences a clinical decision or a robot action that can affect patient safety” [35, 36]. Figure 14 assigns each of the eleven duties the bill creates to the actor who owes it, writing each duty inside the party that owes it rather than beside it, so the six lines that remain in the figure are only the places where two parties have to interact; and Table 14 lists the statutory sections behind those duties.

Section	Duty created	Actor	Prior-law counterpart
Sec. 2	Findings, the evidence to law to cost record	Congress	None
Sec. 3	Amendment to the FD and C Act	Secretary	21 U.S.C. as amended
Sec. 4	Comparative provisions across prior versions	Secretary	Partial
Sec. 5	Financial data, cost ledger per verification run	Sponsor	None
Appendix 6	Cost estimate for the verification regime	Sponsor	None
Appendix 7	Verification economics, per-run pricing	Sponsor	None
Appendix 8	Financial standard for reported data	Secretary	Partial
Appendix 9	Research matrix across prior developments	Secretary	None
Appendix 10	Transparency of the reported ledger	Secretary	Partial

Table 14. The statutory sections, the duty each creates, the actor who owes it, and whether prior law contains anything comparable.

5.2 How the text got there

Figure 15 traces the drafting history, and Table 15 states the same lineage as citable text. The eleven days between version 1.0 and version 5.0 are the quantitative point: in the prior system, five substantive revisions of one bill plus four companion documents is a legislative session's work rather than a fortnight's. Each version's delta is traceable to a named companion. Version 4.0 put machine-readable diagrams inside statutory text, which the narrative case document had argued for; version 5.0 attached a cost ledger to every verification run, which the clinician trust framework and the enactment framework had together made necessary [37, 32, 38].

Version	Deposit	Delta over its predecessor	Companion that supplied it
v1.0	Jun 1, 2026	Standalone Verification Before Generation Act	VVUQ Physical AI Oncology Trial Bill
v2.0	Jun 4, 2026	Recast as an FD and C Act amendment	Carried from v1.0
v3.0	Jun 4, 2026	Findings rendered as figures in statutory text	Carried from v2.0
v4.0	Jun 7, 2026	Machine-readable diagrams inside the bill	From H. R. 9510 to Federal Law
v5.0	Jun 10, 2026	Financial data amendment, cost ledger per run	Earning the Clinician's Trust; Earning the Congress's Vote
Total	Jun 1 to 10	Five versions, four companions	Eleven days

Table 15. The five bill versions, the delta each added, and the companion document that supplied the material, across eleven days in June 2026.

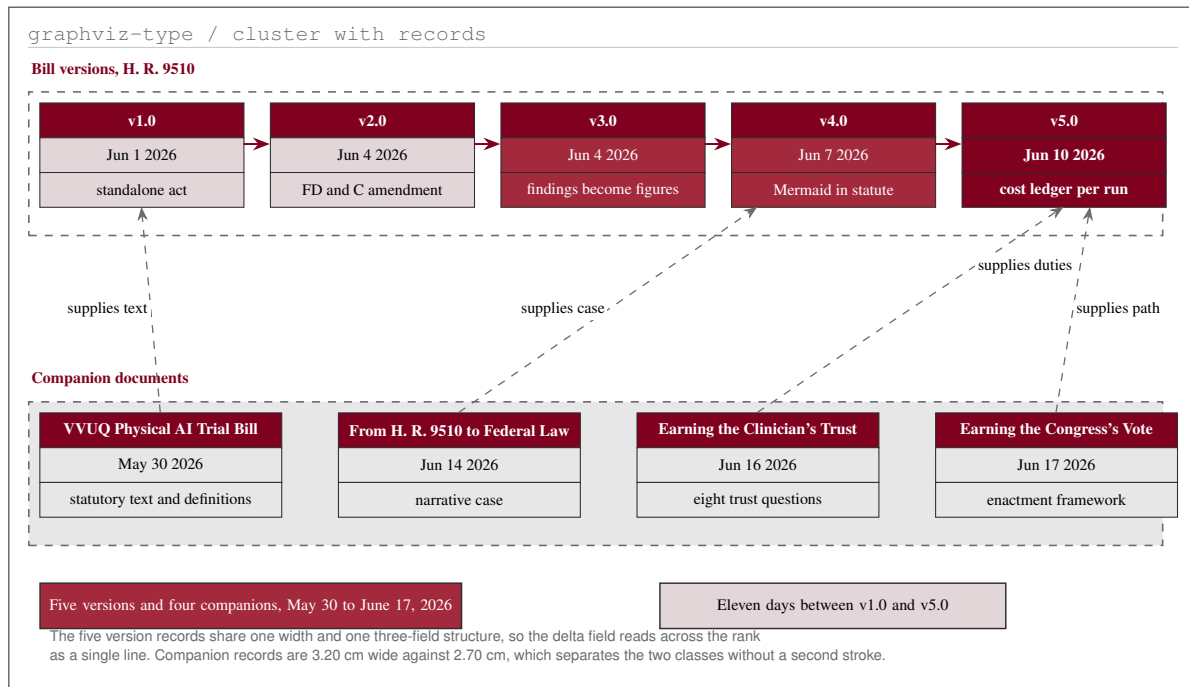


Figure 15. Five bill versions across eleven days, the delta each added, and the four companion documents that supplied the material for those deltas.

5.3 Where the requirement lands

Figure 16 follows one requirement downward through four layers. The statute names it, an adapted 21 CFR Subpart J constrains it, the Phase 0 gate of §4 executes it, and a pre-incision checklist states it in the words a site operator reads. The clinician trust framework supplies the bedside form of the same duty: above the two supervision modes it defines, “sits an unconditional control: the clinician can override and stop the system at any time, for any reason, without satisfying the gates first” [32]. That sentence is what layer 4 of Figure 16 is written to make executable, and it is why the legislative work in this repository is not an abstraction: it terminates in a sentence somebody obeys on the day of surgery. Table 16 places every legislative artifact on the same nineteen-day window.

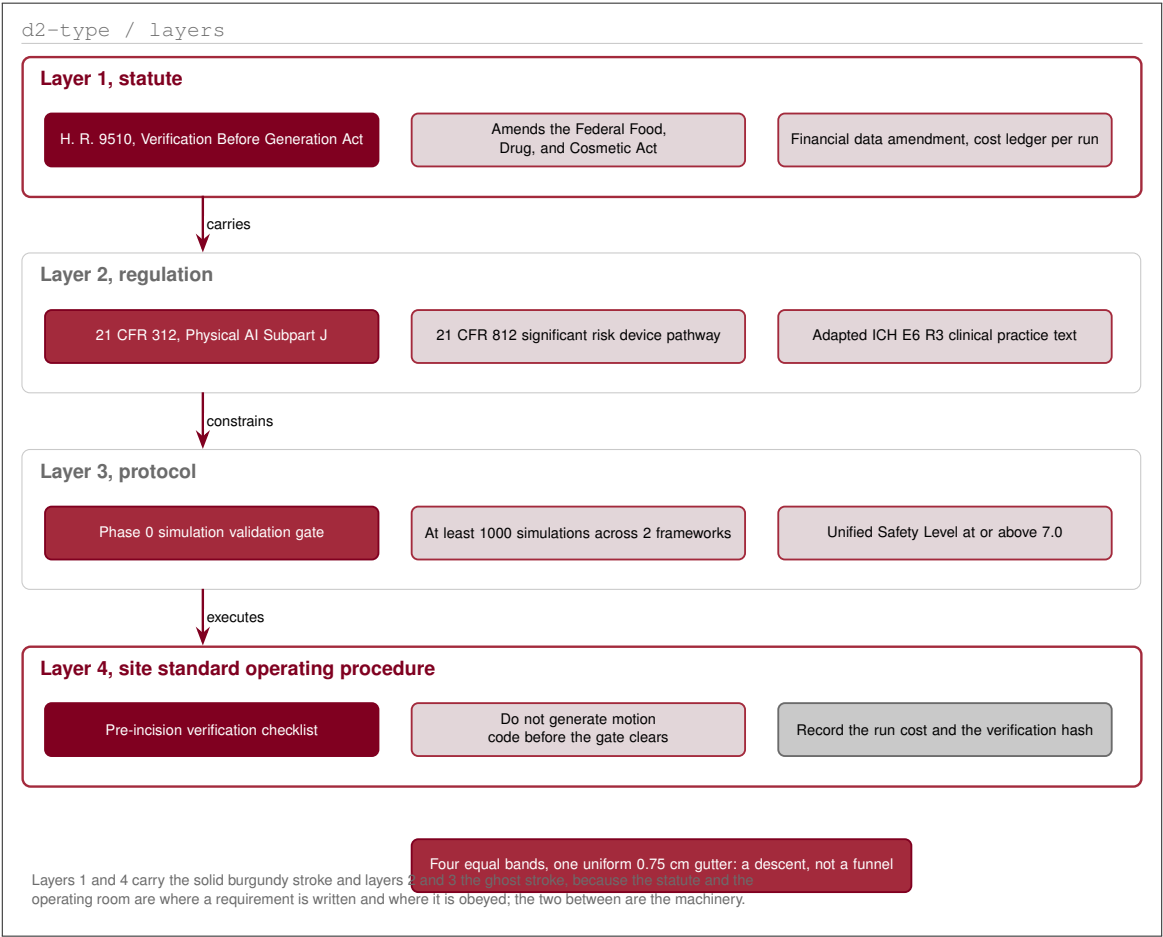


Figure 16. One requirement through four layers, from a statutory sentence to the operating-room instruction, with the artifact that carries it at each step.

Legislative artifact	Deposit	Gap to next	Prior-system counterpart
VVUQ Physical AI Oncology Trial Bill	May 30, 2026	2 days	A committee print cycle
Verification Before Generation Act of 2026	Jun 1, 2026	3 days	A first draft
H. R. 9510 v3.0	Jun 4, 2026	3 days	A markup revision
H. R. 9510 v4.0	Jun 7, 2026	3 days	A markup revision
H. R. 9510 v5.0	Jun 10, 2026	4 days	A reported bill
From H. R. 9510 to Federal Law	Jun 14, 2026	2 days	A committee report
Earning the Clinician’s Trust	Jun 16, 2026	1 day	A stakeholder brief
Earning the Congress’s Vote	Jun 17, 2026	end	A floor memorandum

Table 16. Eight legislative artifacts in nineteen days, the gap between each deposit, and the prior system’s name for the equivalent stage.

5.4 What a bill has to survive

A bill is not a paper, and the difference is worth stating because it governs how the legislative artifacts in this repository were written. A paper is finished when it is deposited. A bill is finished when it survives a markup, a committee report, a cost estimate, and a floor vote, and each of those stages is run by people who did not write it and who are entitled to ask what it costs. The four companion documents exist because each answers one of those questions before it is asked.

From H. R. 9510 to Federal Law answers the narrative question: why this statute rather than an agency guidance, and why a sequencing rule rather than a performance standard [37]. A performance standard says the robot must be safe. A sequencing rule says the verification must clear before the code is generated, which is checkable at a moment in time rather than inferred from an outcome after the fact, and that is the property the embodiment problem makes necessary.

Earning the Clinician's Trust answers the adoption question through eight framed questions, of which oversight and reliability are the two the statute depends on [32]. Its oversight chapter distinguishes two supervision modes: human-in-the-loop, where “the system proposes an action under verification before generation, but it cannot act until the clinician accepts it”, and human-on-the-loop, where “the system acts under audit within pre-authorized bounds while the clinician monitors and retains the power to intervene at any moment”. It then names the failure mode a statute cannot fix by itself: over-trust, or automation bias, “the tendency to defer to a machine even when it is wrong”, against under-trust, “the opposite error of discarding useful guidance”. The design target is calibrated reliance, and the statutory gate exists to make calibration possible rather than to replace it.

Earning the Congress's Vote answers the enactment question, and the financial appendices of version 5.0 answer the cost question directly: a verification regime that cannot be priced per run cannot be scored, and a provision that cannot be scored does not reach a floor [38, 33]. That is why the last substantive change to the bill in eleven days of drafting was not a safety clause but a ledger clause. The three appendices that carry it, cost estimate, verification economics, and financial standard, are the ones a committee staff member reads first and the ones the prior five versions did not have.

5.5 What the legislative work does not claim

Nothing in this repository is enacted, introduced, or endorsed. H. R. 9510 is a drafting exercise carrying a plausible bill number, written to the form of a House bill so that its provisions can be read against the Federal Food, Drug, and Cosmetic Act as they would have to be read if they were ever offered [33]. The companion documents are analyses by a single author. The value of the exercise is not that it will pass. It is that a sponsor preparing a Physical AI oncology trial can now point at a specific statutory sentence, a specific regulatory location, a specific protocol gate, and a specific operating-room instruction, and show that the four agree; and a regulator or a legislative staffer reading the same four can find the disagreement, if there is one, in an afternoon rather than in a comment period. That is the whole of the claim, and Figure 16 is the shape of it.

5.6 Why a sponsor drafted a statute at all

The obvious objection to §5 is that drafting legislation is not a sponsor's job. It is not, and the reason it was done anyway is worth stating because it generalizes. A sponsor preparing a Physical AI oncology trial under existing law discovers that the thing it most needs to promise cannot be promised in any recognized form. It can promise that its device meets a performance specification, because 21 CFR §812 knows what that is. It can promise that its drug is manufactured to a standard, because 21 CFR §312 knows what that is. It cannot promise that the code governing robot-patient contact was verified before it was generated, because no regulation names that ordering and therefore no filing has a place to record it.

Faced with that gap a sponsor has three options. It can ignore the ordering and promise only what the existing categories accept, which is what the prior system does. It can invent a private assurance and ask a site to accept it, which transfers the problem to an institutional review board with no basis for evaluating it. Or it can write what the missing category would say, publish it, and build its own protocol to the standard it proposes, so that the promise is at least legible even where it is not yet required. The third is what this repository did, and the Phase 0 gate of §4 is the same requirement expressed in the one place a sponsor does control.

That is also why the bill's operative provision is a sequencing rule rather than a performance standard. A performance standard would be satisfiable by a sponsor's own testing after the fact, and would therefore add nothing to what 21 CFR §812 already asks. A sequencing rule is satisfiable only by evidence that existed before the code did, which is a different kind of record and one a verification service could be asked to hold. The two duties Figure 14 places in a class no party in force today occupies are the whole content of that difference.

5.7 Where the legislative work is weakest

Three weaknesses are worth conceding. The first is that a verification service does not exist, and a statute that creates a duty no party is positioned to discharge is not self-executing; the bill's appendices price the regime but do not name who would be accredited to run it. The second is that the cost ledger of version 5.0 assumes verification runs are priceable per run, which is true of the author's own simulation record and may not be true of a service operating at national scale. The third is that the eight trust questions of the clinician framework were written by the same author as the bill they inform, so their agreement is not independent evidence [32, 33].

None of the three is fatal to the exercise, because the exercise is not to enact a statute. It is to give a sponsor, a site, and a reviewer one written object that says what verification before generation would mean if it were required, and to build a protocol that satisfies it whether or not it ever is. Table 16's nineteen days measure how quickly that object can now be produced; the three weaknesses measure how much of it would survive contact with a committee, and the honest answer is that the sequencing rule would and the pricing might not.

5.8 How the four documents divide the work

The four legislative artifacts are not four drafts of one thing; each answers a question the others cannot. Setting them side by side is the clearest way to see what a sponsor-side legislative effort has to cover, and it is the reason nineteen days produced four documents rather than one long one.

The VVUQ Physical AI Oncology Trial Bill supplies the definitions. Without it, every operative sentence in H. R. 9510 would have to carry its own vocabulary, and a bill that redefines verification, validation, uncertainty quantification, high-risk artificial intelligence and AI-enabled device software function inside its own text is a bill nobody can read against existing law. Anchoring those terms to ASME V&V 40 and to the device regulations means a reviewer can check each definition against a standard rather than against the author's intent [35, 36].

H. R. 9510 supplies the operative provision and, by version 5.0, the money. Its findings section carries the engineering record the provision rests on, its amendment section places the requirement inside the Federal Food, Drug, and Cosmetic Act rather than beside it, and its financial appendices price the regime per verification run [33]. The placement matters as much as the provision: a freestanding act creates a parallel regime that a sponsor must satisfy twice, while an amendment creates one regime that the existing inspection and enforcement machinery already knows how to reach.

From H. R. 9510 to Federal Law supplies the narrative a committee needs, and *Earning the Clinician's Trust* supplies the bedside form of the duty, which is where a statute either becomes practice or becomes paperwork [37, 32]. *Earning the Congress's Vote* supplies the enactment path [38]. Between them the five

documents cover definition, provision, placement, cost, narrative, practice and path, and Table 16 shows that the whole set was produced in the interval a single markup cycle occupies in the prior system.

One consequence is worth drawing out for a funder rather than for a legislator. A sponsor that can produce this set in nineteen days can also revise it in nineteen days when a regulator, a site, or a committee objects. The prior system's cost of legislative engagement is dominated by the cost of the next draft, and a party that cannot afford the next draft argues its first one longer than it should. That asymmetry is the same one §3 identifies in the documentation branch of the clinical hold fault tree, and it has the same resolution: when re-issue is cheap, a defensible position is one you can revise rather than one you have to defend.

6 Funding Proposals

6.1 Ten applications, one method

In August 2026 ten funding applications were filed for a Phase 1 trial of an LLM-advised robotic pancreaticoduodenectomy with perioperative daraxonrasib, written throughout as an independent scientist [39]. Nine of the ten address mechanisms that fund a person or an institution. One, application 05, addresses the only mechanism in the set built for a company, and it was the shortest of the ten. Each application answers a specific clause: application 01 cites the Director's Pioneer Award as the model for "scaling long-horizon grants for the best and brightest", application 03 addresses "the first federal program designed to fund independent research organizations outside traditional academia", and application 05 answers the sentence that "programs like SBIR open doors for technician-founded ventures" [39, 40]. Table 17 lists all ten with the mechanism class and the ask each carries.

No.	Recipient	Mechanism class	Term	Ask
01	NIH Common Fund, Pioneer Award	Person-based	5 years	\$3.5M
02	ARPA-H mission office	Organization	36 months	\$2.1M
03	NSF TIP Directorate, X-Labs	Organization	5 years	\$3.5M
04	DOE Office of Science, Genesis Mission	Partnership	5 years	\$3.5M
05	NIH SEED, SBIR and STTR	Small business	9 plus 24 months	\$1.6M
06	Foundation for the NIH, AMP	Partnership	5 years	\$3.5M
07	HHMI Investigator Program	Person-based	7 years	\$0.7M per year
08	NCI CTEP	Concept review	5 years	\$3.5M
09	Convergent Research, FRO program	Organization	5 years	\$3.5M
10	UC San Diego Moores Cancer Center	Partnership	One meeting	None

Table 17. The ten applications filed in August 2026, the mechanism class each addresses, the term requested, and the ask carried on each.

Figure 17 places all fourteen 2026 funding artifacts on one calendar, and Figure 20 draws the machinery that produced them. Four of the fourteen closed within one week of the work that fed them, which is the production-rate claim in its most checkable form.

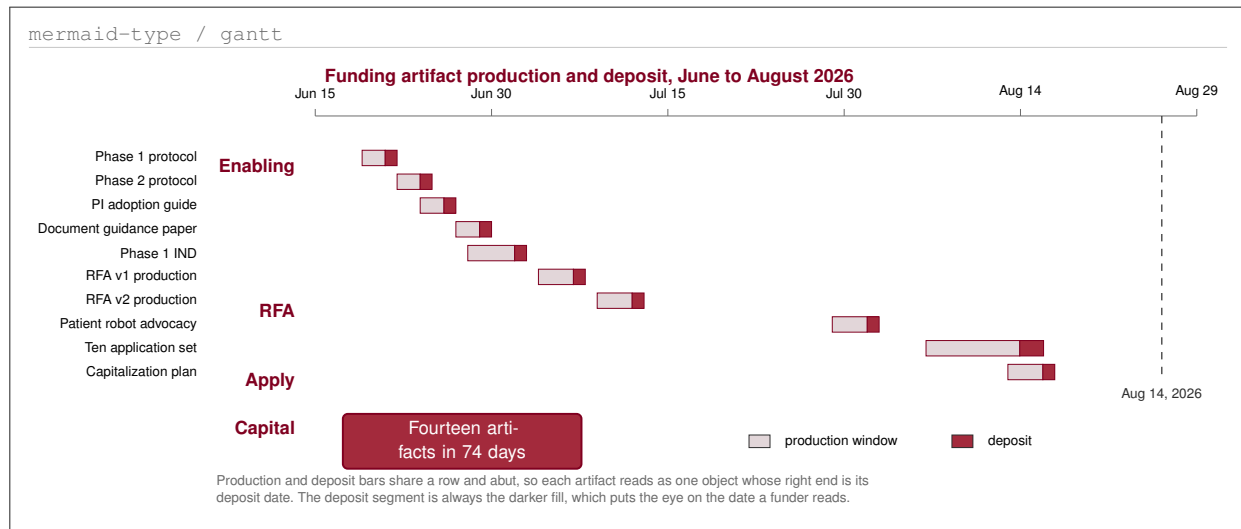


Figure 17. Fourteen funding artifacts deposited between June and August 2026, their production windows, and the four deposits that closed within one week.

6.2 The capitalization plan

The capitalization plan of August 11, 2026 converts the author from independent scientist to small-business operator and rewrites application 05 as the document it should have been [41]. It states its arithmetic without softening: “The identical four-layer budget costs \$3,500,000 of direct work over sixty months and \$1,396,000 of direct work over thirty-three. The award that carries the second is \$1,606,000, a 15.0 percent overhead and fee load against an SBIR allowance of 40 percent that the company does not claim. Routed through a university at a 57 percent facilities and administrative rate, the same direct work would cost a funder \$2,137,000” [41]. And it states the bridge: “Federal money of \$1,606,000 sits below a firewall; contributed drug, operating room and pathology sit unpriced in the middle; and \$5,900,000 of private capital sits above, raised only after a milestone closes. That is 3.67 to one on cash alone” [41, 42]. Table 18 collects the ten quantities the plan turns on; Figure 18 prices the same four-layer budget three ways against a fourth column that routes the identical direct work through an institution, and Figure 19 shows where each port of the award lands and which two work packages it does not reach.

Quantity	Value	What it governs
SBIR Phase I	\$306,000	9 months, 5 milestones
SBIR Phase II	\$1,300,000	24 months, 7 milestones
Total award	\$1,606,000	33 months
Direct work inside the award	\$1,396,000	The work the award pays for
Direct work outside the award	\$2,104,000	The rest of the five-year program
Five-year direct program	\$3,500,000	60 months, direct only
Company overhead and fee load	15.0 percent	Against a 40 percent allowance not claimed
Same direct work, university route	\$2,137,000	At a 57 percent facilities and administrative rate
Private capital above the firewall	\$5,900,000	3.67 to one on cash alone
Staffing at full Phase II	2.6 FTE	The team the plan proposes

Table 18. The ten capitalization quantities the plan turns on, and what each one governs in the proposed thirty-three month program.

d2-type / grid

Cost layer	SBIR Phase I, 9 months	SBIR Phase II, 24 months	Five year direct program	Same work, university route
Simulation and verification	in Phase I scope	carried forward	carried forward	same direct work
Regulatory and protocol	in Phase I scope	carried forward	carried forward	same direct work
Site and device readiness	not in scope	in Phase II scope	carried forward	same direct work
Clinical conduct	not in scope	partial	full	same direct work
Award	306000 dollars	1300000 dollars	3500000 dollars	2137000 dollars
Direct work in the award	part of 1396000	part of 1396000	3500000 dollars	1396000 dollars
Overhead and fee load	15.0 percent	15.0 percent	direct only	57 percent

Federal below a firewall, contributed unpriced, private above: 3.67 to one

2.6 FTE at full Phase II staffing, 33 months

Column 5 is the counterfactual: the same direct work routed through an institution rather than through the company, and it is the only column filled neutral across the money band.

Figure 18. One four-layer budget priced three ways, and the identical direct work set under a 15.0 percent load against a 57 percent rate.

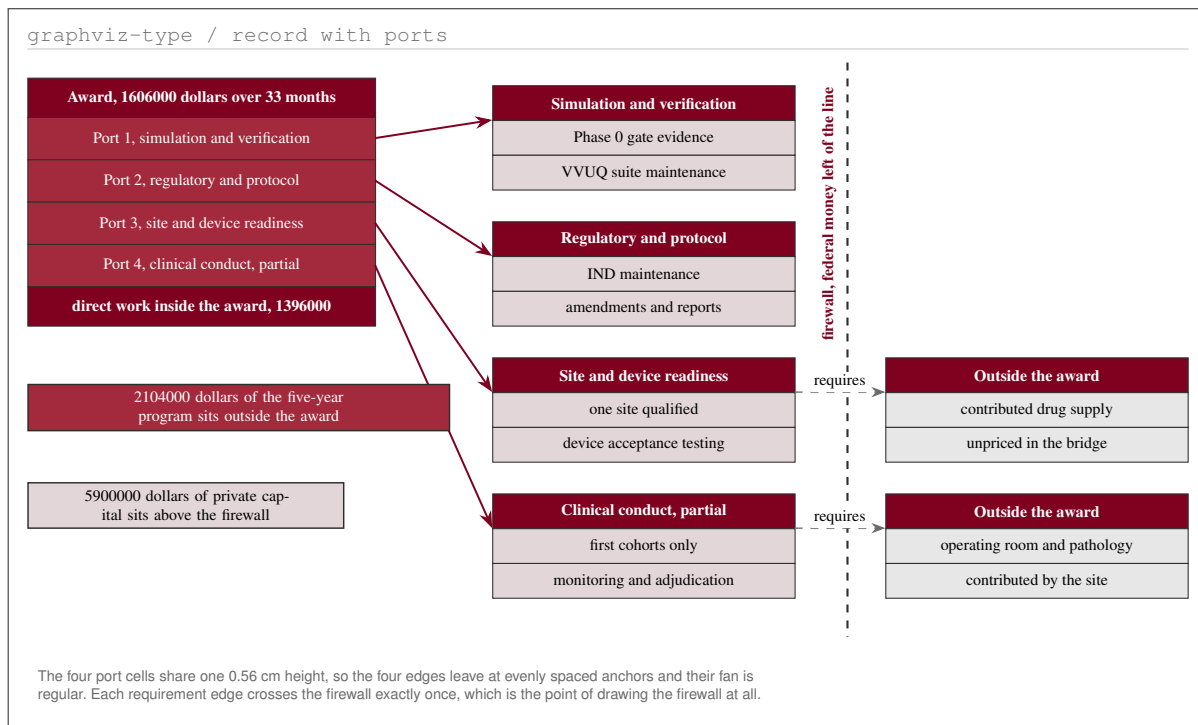


Figure 19. The award as a four-port record, the work package each port pays, and the two packages that sit outside the award and are funded another way.

6.3 The NIH application

Version 2.0 of the RFA-RM-27-001 application, deposited July 12, 2026, is the most complete single funding document in the set: twelve section files carrying the cover and transmittal, opportunity and organization, project summary, research strategy, clinical trial synopsis, human subjects and regulatory, statistics and safety and operations, data management and sharing, facilities and resources, budget and

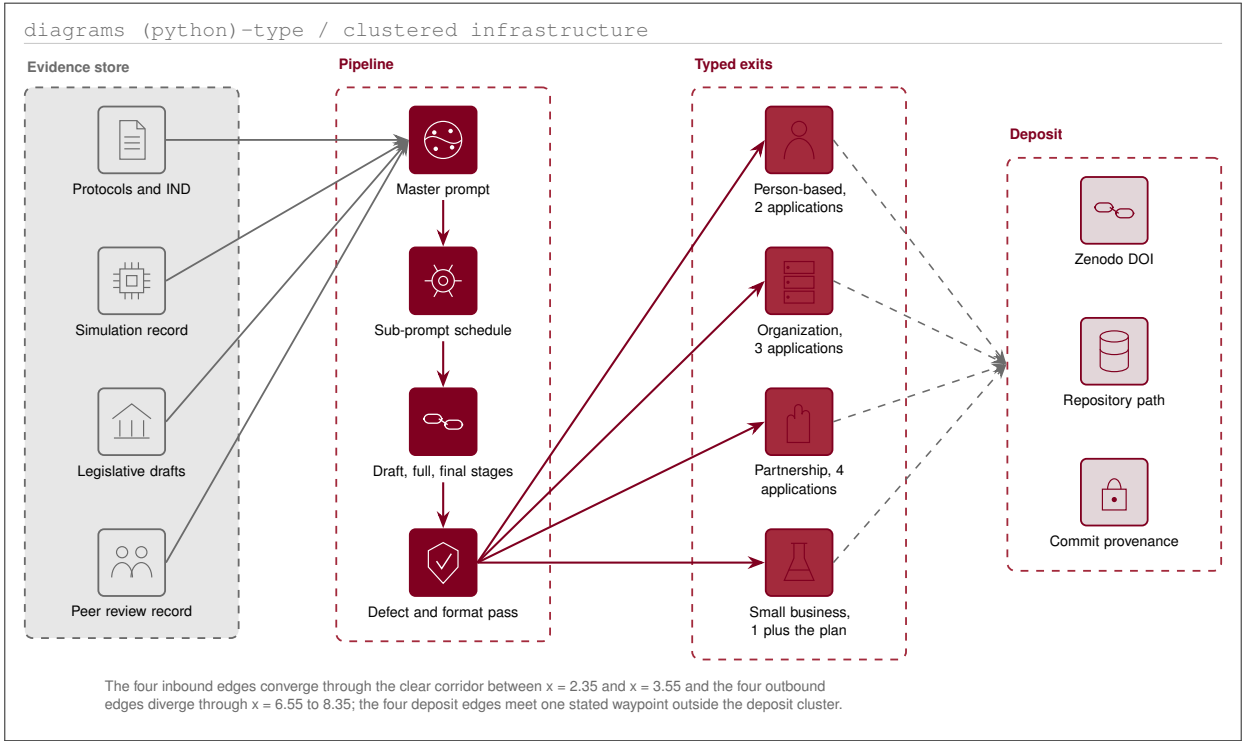


Figure 20. One evidence store, one production pipeline, and the four typed exits at which a funding artifact leaves it for a named recipient class.

milestones and sustainability, investigator profile, and assurances and attachments [31]. Version 1.0 preceded it by five days [43]. Table 19 states what each section had to establish for a reviewer.

No.	Application section	What it had to establish
00	Cover and transmittal	Applicant identity and mechanism fit
01	Opportunity and organization	Why this RFA, and who is applying
02	Project summary	The one-paragraph case
03	Research strategy	Significance, innovation, approach
04	Clinical trial synopsis	The study in the funder’s own form fields
05	Human subjects and regulatory	IRB, consent, and the IND status
06	Statistics, safety and operations	Powering, monitoring, and conduct
07	Data management and sharing	Where the data goes and who may read it
08	Facilities and resources	What exists and what must be obtained
09	Budget, milestones and sustainability	The money and the stop conditions
10	Investigator profile	The single investigator’s record
11	Assurances and attachments	The certifications the form requires

Table 19. The twelve sections of the NIH application at version 2.0, and what each section had to establish for a reviewer.

6.4 What the same work costs each way

The cost case is not an argument about effort; it is a comparison of published benchmarks against a projected figure the author’s own simulation record supports. An empirical virtual trial in triplicate is

benchmarked above \$120,000 per run and about 4.5 months; a ten-arm quantitative systems pharmacology virtual trial above \$2,000,000 and several months to over a year; a digital-twin simulator platform between \$28,000 and \$700,000 and 3 to 16 weeks [41, 44, 45]. Against all three, the author's comparable work is projected at \$36,330 and about one month. The word is projected rather than estimated, because the figure is a projection from the source set rather than a measured invoice, and the capitalization plan reports it that way [41]. Table 20 sets the five activities side by side.

Activity	Industry benchmark	Industry time	This work
Empirical virtual trial, triplicate	Above \$120,000 per run	4.5 months	\$36,330 projected, about one month
Ten-arm QSP virtual trial	Above \$2,000,000	Several months to over a year	\$36,330 projected, about one month
Digital-twin simulator platform	\$28,000 to \$700,000	3 to 16 weeks	\$36,330 projected, about one month
Phase 1 IND dossier	Group effort	Months	4 days, one author
Phase 1 and Phase 2 protocols	Group effort	6 to 12 months each	2 days each

Table 20. Five activities priced against published industry benchmarks, with the projected cost and elapsed time this work reached on each.

6.5 What the applications concede

A funding section that reports only favorable numbers is not useful to a funder, and the capitalization plan does not read that way. It names four concessions in its own risk section, and they are worth restating because a reader of this paper will reach them anyway. The individual-scientist framing that made all ten prior applications distinctive is abandoned when the applicant becomes a company. Capitalization detail the author might prefer to keep private is published permanently under a digital object identifier. Three of the ten prior recipients will read a company plan as off-mechanism. And the plan proposes \$1,606,000 and 2.6 full-time equivalents against a novel-performers chapter that describes work at tens of millions of dollars with teams of ten to a hundred people over half a decade [41, 40].

The last of those four is the one that decides the mechanism. Figure 20's typed exits are not a taxonomy for its own sake: nine of the ten applications address mechanisms that fund a person or an institution, and a firm applying to a person-based mechanism is off-mechanism however good the science is. That is why application 05 and the capitalization plan that rewrites it are the only two artifacts in the set aimed at the exit a company can actually reach.

6.6 What a funder can verify before committing

Every quantity in this section is checkable before any money moves, which is the property the deposit discipline of §2 exists to produce. The ten applications are deposited with their mechanism class and their ask [39]. The capitalization plan is deposited with its four-layer budget, its two overhead regimes, its twelve costed milestones and its stop conditions [41]. Both NIH application versions are deposited five days apart with their section structure intact [43, 31]. The simulation record behind the \$36,330 projection is deposited, and so is the QSP work it derives from [44, 45].

A funder who wants to test the production-rate claim rather than accept it can do so in three steps: read Figure 17's calendar, open the commit history of the directory named on each row, and compare the dates. A funder who wants to test the cost claim can read the benchmark column of Table 20 against the cited sources

and then read the author's own simulation record. Neither test requires the author's cooperation, and neither is available for a plan that exists only as a narrative.

6.7 What the money is actually for

A funder reading Table 18 is entitled to ask what the \$1,606,000 buys that the existing artifacts do not already supply, given that the IND, both protocols and the statutory framework are written and deposited. The answer is in Figure 19's four ports, and none of the four is document production. Port 1 buys the Phase 0 gate evidence: at least 1000 simulated procedures across at least two independent frameworks, and the maintenance of the verification suite that produces them. Port 2 buys IND maintenance, amendments and required reports, which is the recurring regulatory cost of holding an active application rather than the one-time cost of writing one. Port 3 buys one qualified site and device acceptance testing. Port 4 buys the first clinical cohorts with monitoring and adjudication [41].

Document production, the thing this paper is about, appears nowhere in the budget. That is deliberate and it is the strongest statement the funding section can make: the method of §2 has already removed the cost it was built to remove, and the remaining \$1,606,000 is the irreducible cost of running a trial rather than of describing one. A funder who believes the production-rate claims of §3 through §5 should therefore expect the ask to be smaller than a comparable program's, and it is: \$1,606,000 over thirty-three months against a \$3,500,000 five-year direct program, with 15.0 percent overhead and fee against a 40 percent allowance not claimed [41].

6.8 The two prices, read as one sentence

The arithmetic of Figure 18 reduces to a single comparison a funder can carry away. The identical four-layer budget of direct work is \$1,396,000 when routed through a company with no negotiated indirect-cost rate and \$2,137,000 when routed through a university at a 57 percent facilities and administrative rate [41]. The difference, \$741,000, is not a saving on the work; the work is the same work. It is the price of the route. Nine of the ten applications in Table 17 address mechanisms that take the second route, and the capitalization plan and application 05 exist to make the first route available for the same science.

Above that sits the bridge. Federal money of \$1,606,000 sits below a firewall; contributed drug supply, operating room time and pathology sit unpriced in the middle, which is why Figure 19 draws them outside the award and connected to it by requirement edges rather than by port edges; and \$5,900,000 of private capital sits above, raised only after a milestone closes, giving 3.67 to one on cash alone against an annexed target of at least 3 to 1 [41, 42]. A funder is therefore not being asked to fund a program. It is being asked to fund the gate that unlocks one, and the twelve costed milestones state in advance what closing that gate looks like. The economics of the review round belong in the same ledger. Fourteen funding artifacts were produced across 74 days at an inference cost measured in tens of dollars, against list article processing charges of \$2,000 to \$6,000 for a single paper and above \$10,000 at several flagship titles [30]. That asymmetry is not a footnote to the capitalization argument; it is part of it, because every dollar not spent on placing a manuscript is a dollar available to the direct program, and every week not spent waiting on a round is a week of the twelve costed milestones. §7 gives both figures and the ratio between them.

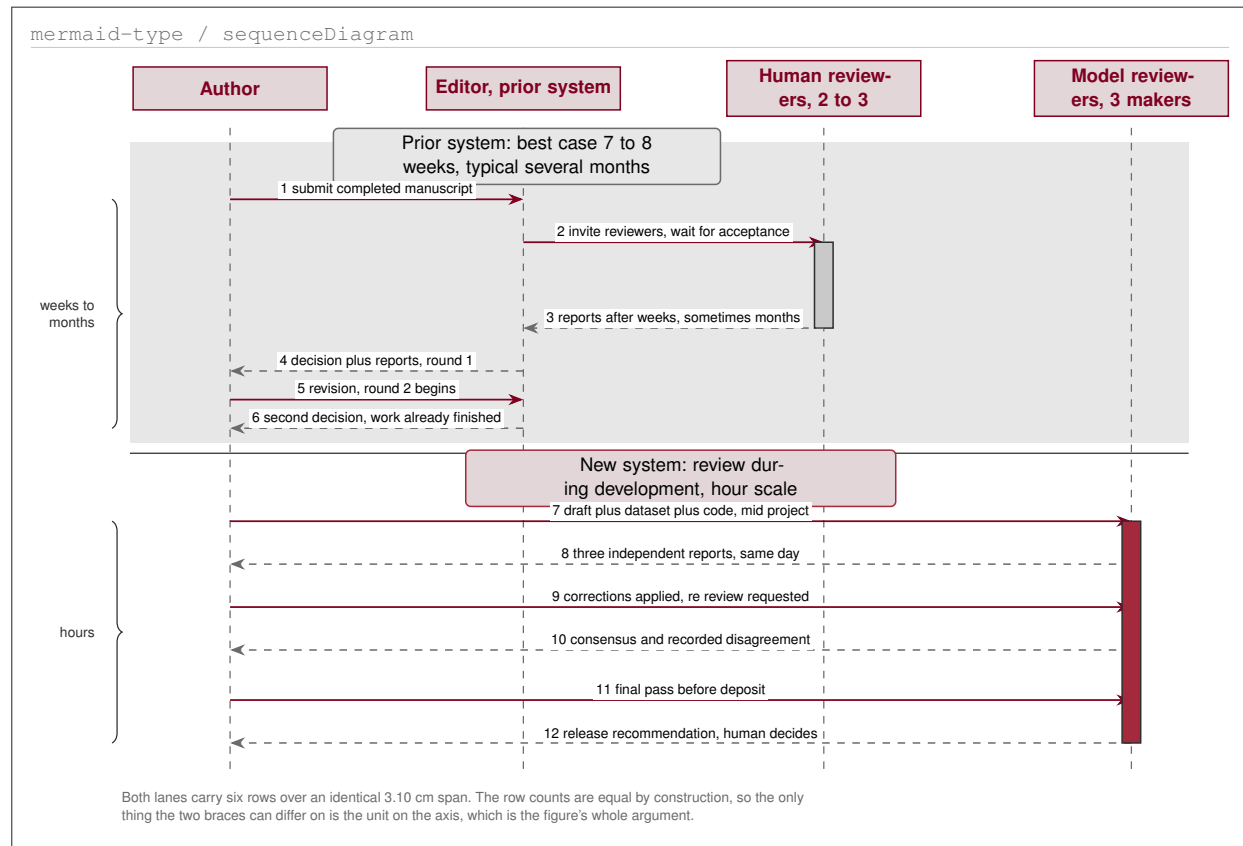


Figure 21. One manuscript on two clocks: the prior system's serial human rounds over months, and the new system's parallel model rounds within hours.

7 AI Peer Review

7.1 The study the method comes from

The AI peer review method used throughout this repository was established in a study deposited November 30, 2025 [30]. That study was about clinical trial patient matching in glioblastoma, not pancreatic cancer. The disease is not what transfers; the review method is, and it transfers without modification to the pancreatic cancer work reported in §3 through §6. The study states its own central claim in one sentence: “A primary benefit of AI peer review is the ability to appropriately conduct iterative analyses and corrections throughout the entire manuscript process. This responsible method improves on over-worked and fatigued human recommendations provided after manuscript completion” [30]. Everything else in this section is that claim's measurement. The prior system's timings are the study's own: the late 1990s and 2000s brought electronic mail and web portals “to reduce physical mail delays by up to weeks, for a best-case review cycle of 7-8 weeks”, and an average journal publication today “can still take several months for processing, including 1-2 rounds of review”, with faster online journals at about one month [30]. Figure 21 lays those timings against the new system's on one clock.

7.2 What the delay costs

Speed is the weakest form of this argument, because a reader cannot act on an adjective. The four costs the prior regime imposes can be stated as quantities instead, and each is being paid now rather than in principle.

The first cost is measured in patients. Pancreatic cancer is the third leading cause of cancer death in the United States, and the 2025 statistics put US pancreatic cancer deaths at about 51,980 for the year, against a total five-year survival below 13 percent [24]. That is about 142 deaths a day and about 1,000 a week. A

best-case review round in the prior regime is 7 to 8 weeks, or about 52 days, so roughly 7,400 Americans die of this disease inside one best-case round; a typical processing time of several months covers 17,000 to 26,000. Peer review does not cause those deaths. What the arithmetic establishes is narrower: the latency of a review regime is a clinical quantity for this disease, not an administrative one. A dossier, a protocol amendment or a funding decision that clears in a day rather than in a quarter reaches a different set of living patients, and that set is countable.

The second cost is measured in what a paper is allowed to show. The prior regime holds a research article to six or eight figures and tables combined, for a reason that was once good: every figure cost a person hours to draw and a reviewer minutes to check. This paper carries twenty-five figures and twenty-five tables, because each figure here is a specification file rather than an image: twenty-five files, about 158 KB in total and about 6.5 KB each, every one carrying a perspective statement, valid source in its platform's own syntax, an absolute coordinate construction table and an edge-routing argument, as §2 sets out. A reviewer who doubts Figure 14 reads its coordinates rather than measuring a rendered image. Substantial, quickly generated visual evidence is what the prior regime is least equipped to accept, and it is what a funder asks for first.

The third cost is measured in dollars and in months of lost research. List article processing charges commonly run from about \$2,000 to \$6,000 per article and exceed \$10,000 at several flagship titles, and the researcher pays them. The comparison figure from the source study is exact: its evaluation standard scored a perfect project completion against " $n_p = 5$ prompts within $n_d = 12$ days while only spending $n_c = \$35$ " [30]. Thirty-five dollars of inference carried a fourteen-day study through a dataset, two notebooks, six regulatory documents and a working API repository. The months matter more than the dollars. An author placing three papers a year in the prior regime spends twenty-one to twenty-four weeks a year waiting on rounds, and the source study names what those weeks are: the new regime "eliminates publication lull and cognitive load of uncertainty, as the researcher simply transitions to the next AI peer reviewed project" [30].

The fourth cost is the direction the money runs. Under the author-pays model the researcher transfers the charge to the journal, and what the journal supplies in return is a page on its own site carrying the researcher's work, with the indexing, the traffic and the promotion of that work accruing to the journal's own title. The source study states the alternative: authors "can now improve their speed of development and gain reputation in papers by simply providing prompts, LLMs, AI peer review recommendations, and corrections they made - without the need for paid review or journals", and "interested parties can then view the author's peer review findings alongside the paper" [30]. Every artifact in this repository is deposited to a DOI and to a public repository at no charge to the author, under CC BY, with the review findings deposited beside the work rather than withheld from it. Table 21 puts the four costs beside what replaces each of them, and Figure 22 computes the ratio.

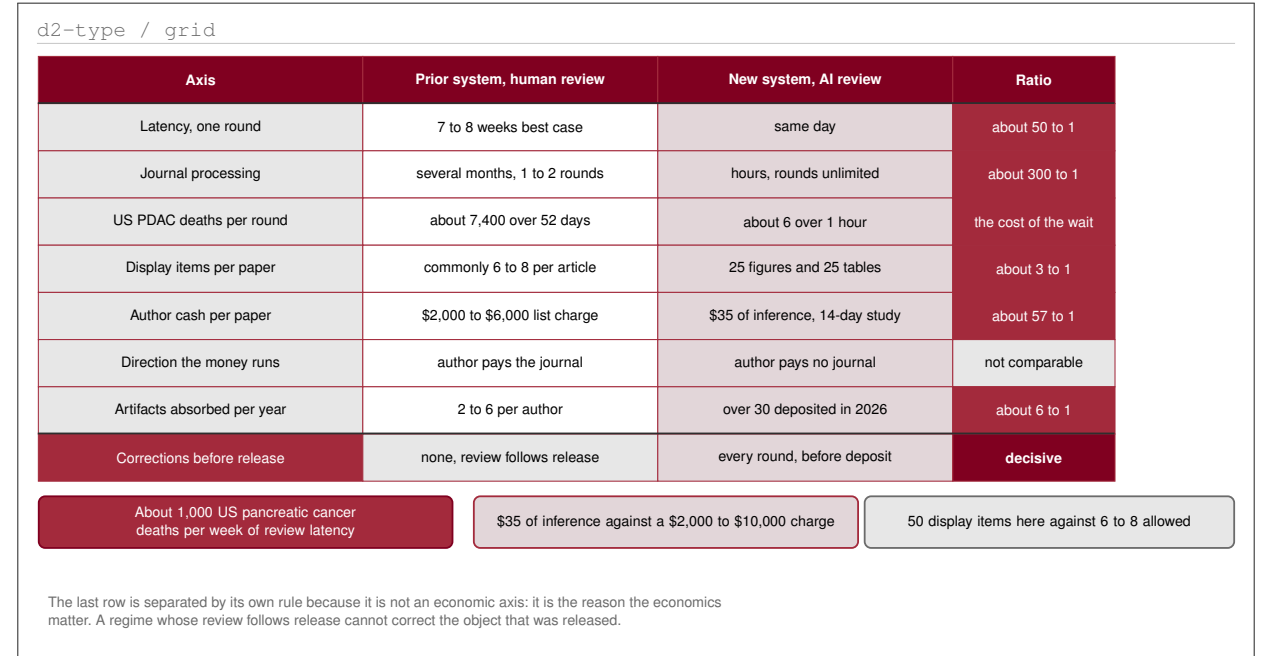


Figure 22. Seven measurable axes of one review round under both regimes, with the ratio in each cell and the decisive row cut away below them.

Axis	Prior system	New system	Ratio or delta
Latency, one round	7 to 8 weeks best case	Same day	About 50 to 1
Typical journal processing	Several months, 1 to 2 rounds	Hours, rounds unlimited	About 300 to 1
US pancreatic cancer deaths inside one round	About 7,400 over 52 days	About 6 over 1 hour	The cost of the wait
Display items one paper carries	Commonly 6 to 8 per article	25 figures and 25 tables	About 3 to 1
Author cash cost per paper	\$2,000 to \$6,000 list charge, over \$10,000 at flagship titles	\$35 of inference across a 14-day study	About 57 to 1 and up
Direction the money runs	Author pays the journal to host the author’s work	Author pays no journal, deposits under CC BY	Not comparable
Reviewers per round	2 to 3 humans	3 model manufacturers	1 to 1 by count
Artifacts absorbed per year	2 to 6 per author	Over 30 deposited in 2026	About 6 to 1
Corrections before release	None, review follows release	Every round, before deposit	The decisive axis

Table 21. Eight measurable axes of one review round under both regimes, and the ninth axis on which the two regimes are not comparable at all.

Figure 22 carries seven of those axes as a grid, with the ratio computed in each cell and the decisive row separated by its own rule. A regime whose review arrives after release cannot apply a correction to the released object, whatever its cost or its latency, and that is why the last row is not an economic axis at all.

7.3 What three days buys

Three things follow from the four costs, and each is stated here as a claim with a date attached rather than as a prospect.

This paper was finished within three days. The master prompt was supplied on August 12, 2026 and this document was deposited on August 14, 2026, with twenty-five figures, twenty-five tables, eleven section files and about 324,000 characters of LaTeX source, across an eight-stage build whose every file was committed the moment it was written. Nothing in that window waited on anyone. The point is not that three days beats several months; it is that the prior regime has no state in which a document of this size and this figure count exists at all before the first review round has closed.

Major revisions run on the 1 to 2 day scale. This version of the paper is the evidence for that claim rather than an assertion of it. A second author prompt on August 13, 2026 required Figure 14 to be redrawn from a use case diagram to a class diagram, this section to be rewritten around quantities, Table 21 and Figure 22 to be re-cut to the same axes, and a defect that stopped the prior stage compiling at all to be found and fixed; the result was deposited the following day. In the prior regime a change of that size is a revision round, and a revision round is 7 to 8 weeks at best. The new regime completes about thirty of them in the time the prior regime completes one, and completes them before release rather than after it.

On this record, human peer review is not required for this class of work. That is the author’s position and it is stated plainly, because hedging it would misrepresent how the repository is actually run. The basis is the work from late 2025 to the present: five separately deposited and separately dated review instances, listed in Table 24, whose corrections are visible in the diffs; frontier model results on academic reasoning, scientific knowledge and multimodal benchmarks that improved across every release in that window [30]; and a production rate that the prior regime’s throughput cannot meet at any level of goodwill. The position is bounded, and the three bounds are set out below under what AI peer review is not: it concerns journal peer review of the author’s own deposited works. It does not touch institutional review board approval, an FDA information request, a data and safety monitoring board, or any other regulatory determination, none of which is a peer review process and none of which this system proposes to replace.

7.4 Three manufacturers, one human gate

The pancreatic cancer funding application makes the role assignment explicit and binding. Its provenance section states that the “planned tripartisan workflow assigns Claude Code the primary production role for code, regulatory adaptations, protocols, IND materials, diagrams, and repository-scale artifacts. ChatGPT Codex serves as the first independent peer reviewer, emphasizing verification, validation, uncertainty quantification, external standards, trial-literature synthesis, deep research, and PDF review. Google Gemini serves as the second independent reviewer, emphasizing accelerated code problem solving, meta-verification of prior stages, and administrative workflow support” [31]. The reviews “occur at defined mid-project milestones and after material corrections so that findings can change the work before release, with the human PI retaining final authority”, and the same paragraph records that these systems “are disclosed as drafting and peer-review aids” [31]. Table 22 names the three manufacturers, the role each holds, and the human principal investigator who retains final authority over all three; Figure 23 draws where the concurrency sits and where it stops. Three manufacturers read one frozen artifact and its recorded hash, so no branch can observe another’s output and a disagreement is evidence rather than an artifact of ordering.

Manufacturer	Role	What that role covers in this work
Anthropic	Primary production	Code, regulatory adaptations, protocols, IND materials, diagrams, repository-scale artifacts
OpenAI	First independent reviewer	Verification, validation, uncertainty quantification, external standards, trial-literature synthesis, deep research, PDF review
Google	Second independent reviewer	Accelerated code problem solving, meta-verification of prior stages, administrative workflow support
Human PI	Final authority	Source verification, regulatory accuracy, scientific interpretation, participant protection, final submission

Table 22. The three model manufacturers and the human principal investigator, and the role each of them holds in the reviewed work.

7.5 What the review actually found

The glioblastoma study is worth reporting in its numbers, because the numbers are what make the method transferable rather than aspirational. A ten-file dataset and a clinical BERT notebook reached “a final notebook at a low test set performance of 67.3%”. A triple review by three models from three manufacturers “all yielded the primary recommendation to upgrade to a tabular model, which was more suitable for the csv dataset”; one of them identified an open-source TabPFN-v2 classifier, and the recommendations were implemented to yield “a test set accuracy improvement to 94.0% with few-shot learning along with peer review corrections” [30]. An evaluation standard generated for the purpose scored the workflow at 87.5 percent overall quality and 0.831 efficiency. Eleven attempts to produce a working FastAPI repository failed before a two-model prompting strategy succeeded. Three independent regulatory analysts scored the six generated FDA, ICH and ISO documents at 8.0, 9.1 and 9.1 out of 10, and the study records the reason each gave rather than averaging them [30]. Table 23 collects the measures.

Measure	Value	What changed it
Test-set accuracy before review	67.3 percent	Initial clinical BERT pipeline
Test-set accuracy after review	94.0 percent	Tabular model plus few-shot learning
Overall workflow quality	87.5 percent	Evaluation standard, eight criteria
Efficiency score	0.831	Human, prompt, time and cost metrics
Regulatory analyst scores	8.0, 9.1, 9.1 of 10	Three manufacturers, recorded separately
Failed API attempts before success	11	Resolved by a two-model prompting strategy
Study duration, one author	14 days	Dataset through FastAPI repository

Table 23. What the AI peer review changed, measured, in the glioblastoma study whose method the pancreatic cancer work in this paper inherits.

Figure 24 is the reason those three regulatory scores are reported separately rather than as a mean of 8.7. None of its five terminal dispositions averages the reports, only one leaves a disagreement without an action, and two stop the artifact from progressing at all.

7.6 Why only AI reviewers can carry this load

The study’s conclusion is not that AI review is convenient. It is that the volume, complexity and frequency of AI-generated research have already passed what human review can absorb: this next evolution of peer review marks “the pivot away from slow and irresponsible use of human peer review for LLM-generated

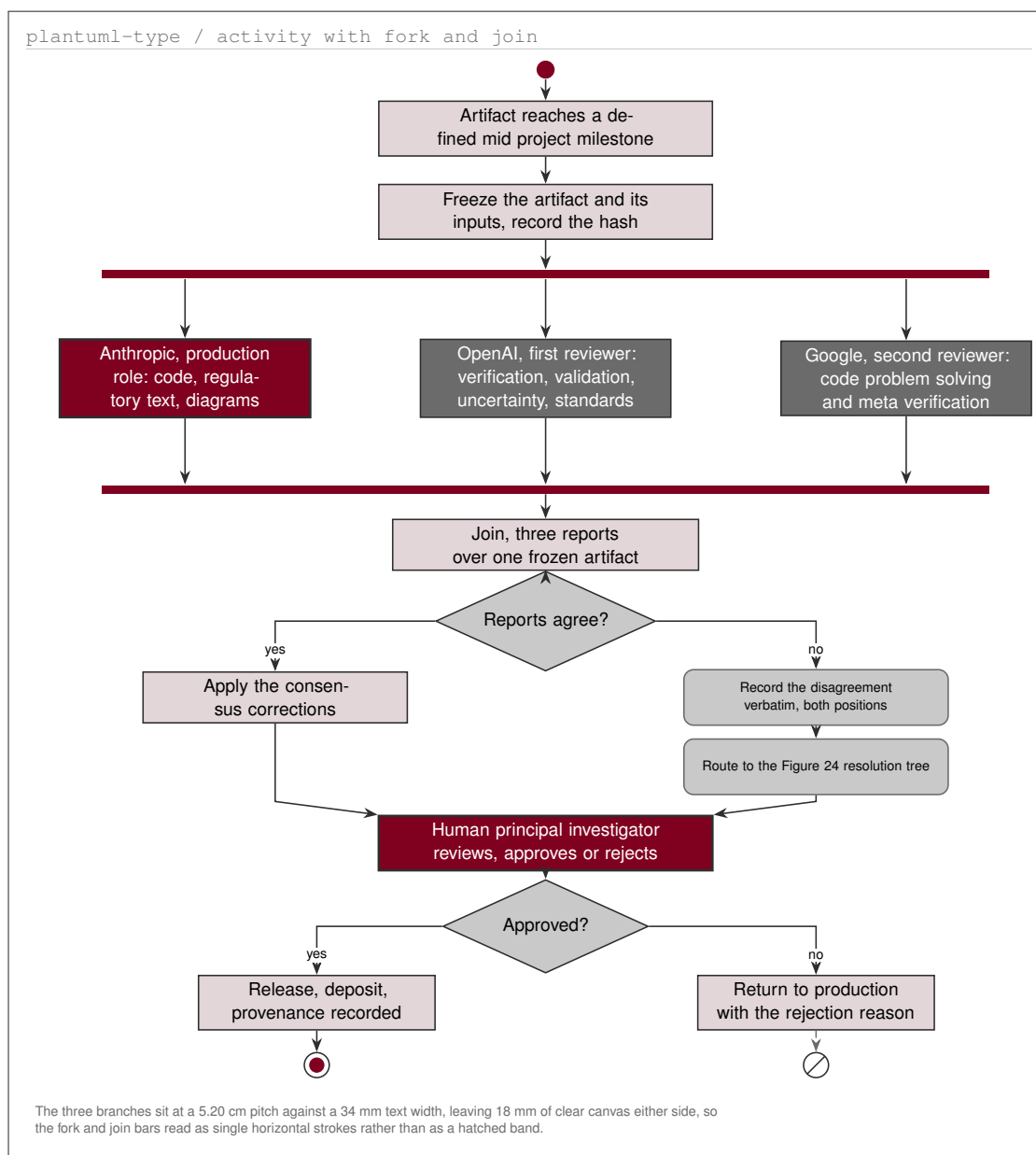


Figure 23. One artifact reviewed concurrently by three model manufacturers, joined into one recorded disagreement set, and cleared only by the human PI.

data, and back to fast automation of patient health applications at scale” [30]. The same study records that trust for the process “was established over a series of human-AI works reflected in LLM code and non-code benchmarking, applications in literature, and prior author works using AI in review roles coordinated across judging, external validation, cross-verification, meta-verification, and tests according to FDA/ICH guidance” [30, 46, 47]. Table 24 lists five instances of that process across the author’s deposited works, and the specific change each produced. The count itself is the argument: in 2024 the author deposited eight works, in 2025 twelve, and in 2026 through August more than twenty, including an IND, two clinical trial protocols, five bill versions, ten funding applications, two NIH application versions and a capitalization plan [22]. A regime that took seven to eight weeks per round in its best case was already the binding constraint at eight works a year; at thirty it is not a constraint but an exclusion, and the work simply does not enter it. That is the sense in which the two systems are incompatible rather than merely different. Human review is not worse; it is

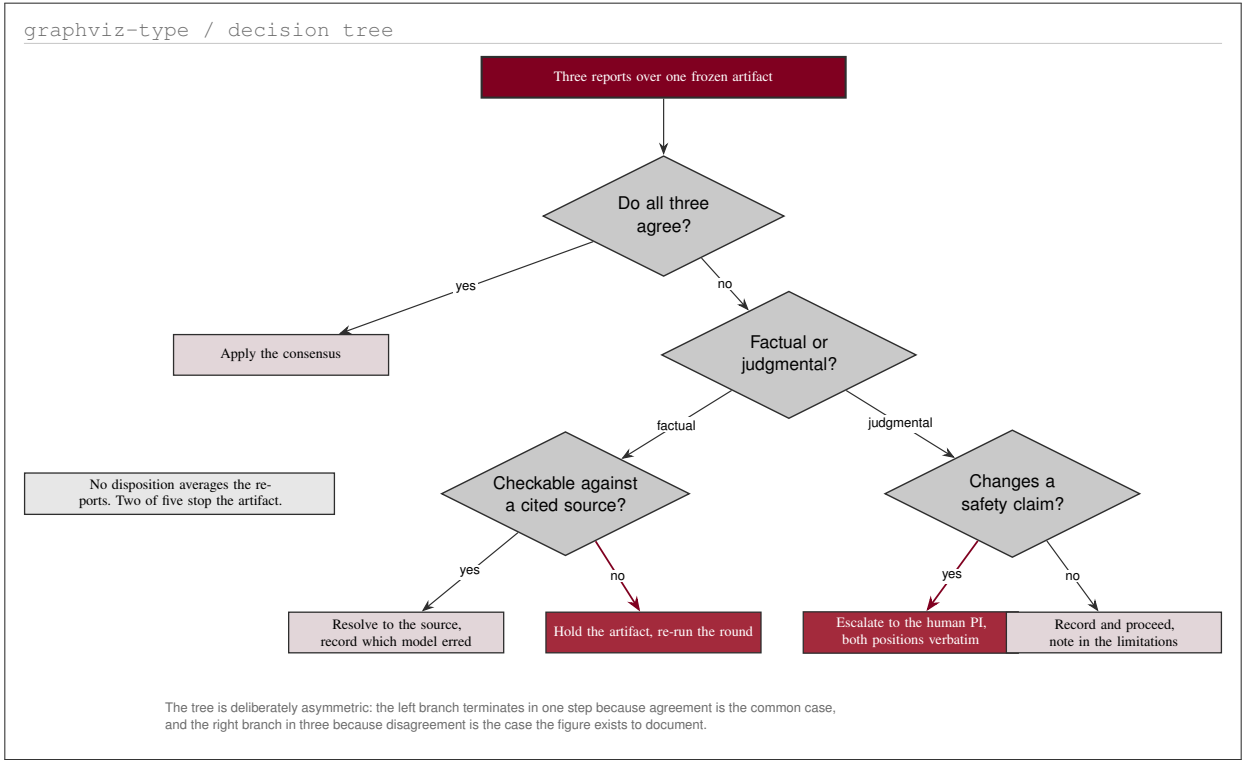


Figure 24. One disagreement from three model reports to five terminal dispositions, of which none is an average and only one is silent.

sized for a production regime that no longer exists on the author’s side of the transaction, and no amount of goodwill inside the prior regime changes its throughput.

Work	Date	Reviewers	What the review changed
Glioblastoma patient matching ML	Nov 30, 2025	3 manufacturers	Model class, 67.3 to 94.0 percent
Code generation competition	Dec 22, 2025	16 models	Iterative learning on FDA adverse events
NIH application v1.0	Jul 7, 2026	2 reviewers	Structure and mechanism fit
NIH application v2.0	Jul 12, 2026	2 reviewers	Budget, milestones, review schedule
Capitalization plan	Aug 11, 2026	2 reviewers	Column widths, caption balance, figure centering

Table 24. Five instances of AI peer review across the author’s deposited works, and the specific change each review produced.

7.7 What AI peer review is not

Three limits bound the position taken above, and the case is weaker if they are left implicit. First, model review is not regulatory review. Nothing in §7 substitutes for an institutional review board, an FDA information request, or a data and safety monitoring board, and the funding application says so in its own words: the models “are not applicants, investigators, sponsors, regulators, clinicians, or decision-makers” [31]. Second, model review is not clinical validation. Improvements over time on coding, scientific-reasoning and multimodal benchmarks support current frontier models as independent reviewers, “but not as clinical validators” [31, 30]. Third, model review does not establish correctness by agreement.

Three models agreeing is three reports, not a proof, which is why Figure 24's tree never averages and why two of its five dispositions stop the artifact rather than passing it. The source study bounds itself the same way, recording that its own evaluation standard was "only based on the current project" and its efficiency metrics "were partly influenced by the author's priorities" [30]; that is why Table 23 reports three regulatory scores separately rather than a single figure of merit.

7.8 The trust that had to be earned first

AI peer review is only defensible if the reviewers were shown to be competent before they were relied upon, and that record runs from late 2025 to the present. In October 2025 an end-to-end oncology trial efficiency study established that LLM-generated regulatory text could be produced against FDA and ICH requirements and checked [46]; in November 2025 a drug synergy machine learning study took rapid code prototypes through to project deliverables [47]; later that month the glioblastoma patient matching study introduced the triple review this section reports [30]; and in December 2025 a code generation competition ran sixteen proprietary and open-source models against FDA adverse event data with iterative learning, which measures reviewer competence rather than assuming it [48]. By the time the pancreatic cancer work of §3 through §6 began in 2026, the question was no longer whether the models could review; it was which model should hold which role, and the answer is Table 22.

The chronology matters because it is the difference between an appeal to authority and an appeal to record. The claim is not that frontier models are reliable in general, but that these models, in these roles, on this class of artifact, produced measurable corrections on five separately deposited and separately dated occasions whose diffs are public. A reviewer who doubts the claim can read the corrections rather than the argument.

7.9 What a human still does

Four things in this system are done by a person and cannot be delegated. The person decides what to build: no model chose to write an IND, two protocols, a bill and fourteen funding artifacts. The person supplies the master prompt and therefore the constraints, including the ones that make the output checkable, such as one file per commit and no raster figures. The person adjudicates disagreement: Figure 24's escalate branch terminates at the principal investigator, and two of its five dispositions stop the artifact. And the person signs, which the funding application states in the plainest terms available, that the models "are not applicants, investigators, sponsors, regulators, clinicians, or decision-makers" [31].

That division is the reason the system can be audited at all. A regime in which a model both produced and cleared the work would have no point at which responsibility attaches, and no funder should accept one. The regime described here attaches responsibility at exactly one point, and the whole of §7 argues that the point is correctly placed.

8 Limitations and Future Work

8.1 Limitations

Two limitations are structural rather than incidental. The first is that a single model family, Claude Code Opus 5, wrote every line of source in this paper and in the eight-stage build that produced it. A defect in that model’s idiom appears in every artifact at once, and no amount of internal consistency checking inside one family detects a shared blind spot. The source study for §7 records the same class of limit from the other direction: its first working notebook used a model whose parameters were “nearly 6 years” old and a newer alternative “may have been a better replacement”, which the author attributes to the generating model relying “on older information from pre-trained parameters as opposed to a more exhaustive online model search process” [30]. It also records a scale ceiling: “there was an upper limit when working with over 1500 lines of code”, and truncation appeared “when including additional documents beyond the notebook for consideration”, so that “the most ideal responses were typically received when the author defined the problem the most directly with the least number of file attachments” [30]. The mitigation is the two independent reviewer manufacturers of §7, and the residual is the case where all three agree and all three are wrong.

The second limitation is text watermarking. Generated text may carry a provenance watermark, detection policy is not settled, and journals differ on what disclosure they require. This paper’s answer is not to argue about detection. It is to disclose the model on the cover page of every deposited work, to make the commit history rather than a claim of authorship the primary record, and to deposit to a repository and a preprint rather than through a journal gate. Figure 25 pairs each exposure with the mitigation that reduces it, and Table 25 states the same register in citable text with the residual that survives each.

Two further limits deserve statement even though they are smaller. This paper reports on documents rather than on patients: no participant has been enrolled, no operation has been performed under these protocols, and every clinical number in §3 and §4 is either a published external result or a planned design parameter. And the review regime of §7 was measured on a glioblastoma machine learning workflow rather than on a regulatory dossier; the method transfers, and the measurement does not. Both limits are removable only by doing the trial, which is what the funding is for.

Exposure	Mitigation	Residual after mitigation
One model family writes all source	Two independent reviewer manufacturers at defined milestones	A defect all three share would survive
One context window bounds each stage	Machine-readable, portable specifications	Re-execution costs a stage, not a project
Vendor availability is a project risk	Every artifact deposited under a DOI as produced	Deposited work is unaffected
Generated text may carry a watermark	Provenance disclosed on every cover page	Disclosure does not remove the watermark
Detection policy is not settled	Commit history is the primary record	Policy may change after deposit
Journals differ on disclosure	Deposit to preprint and repository	A funder requiring journal review is not reached

Table 25. Six exposures, the mitigation applied to each, and the residual that survives it, ending in one funding decision risk.

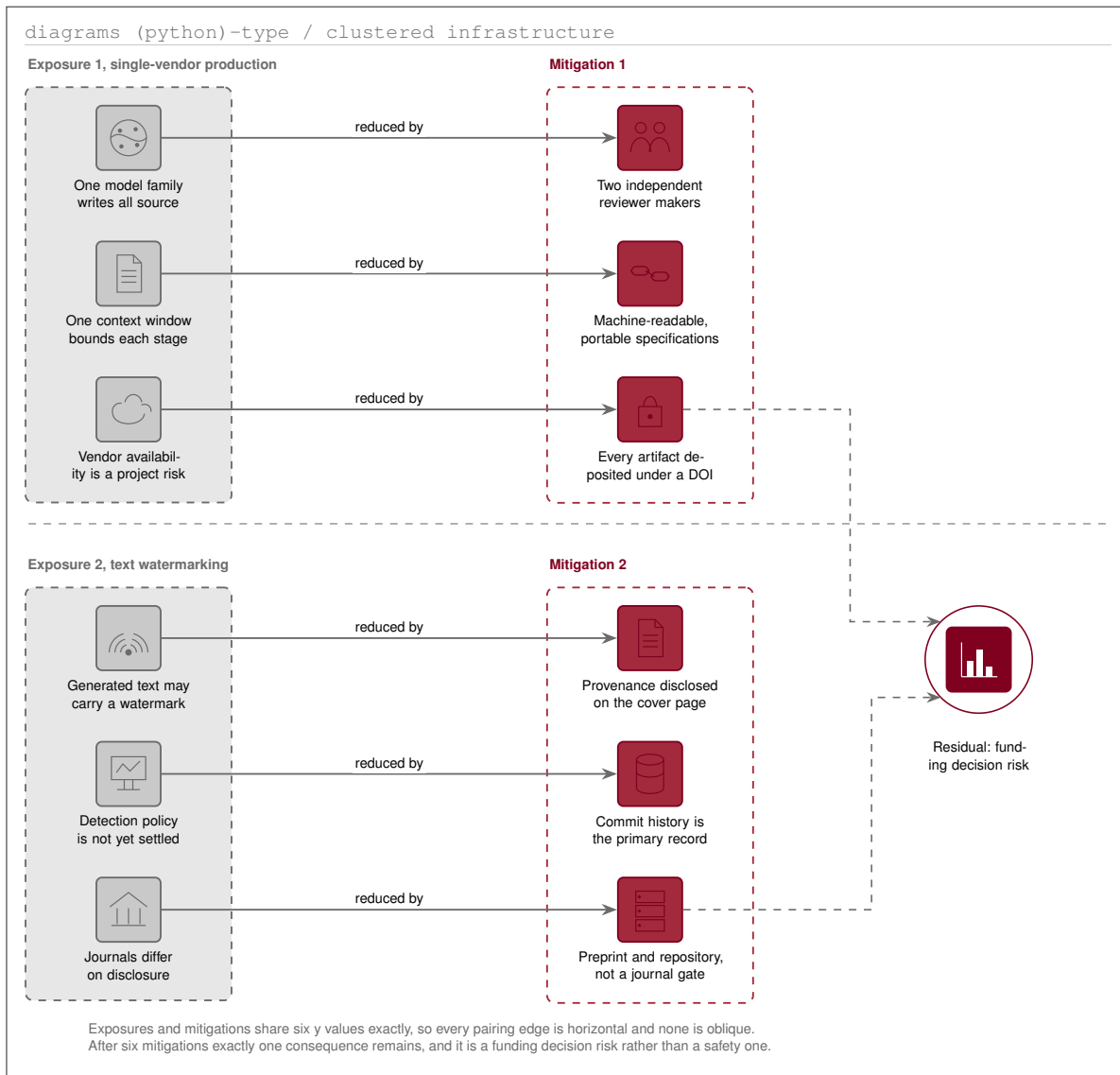


Figure 25. Two structural exposures, single-vendor production and text watermarking, each with the mitigation that reduces it and its residual.

8.2 Future work

Future work is one thing: securing funding for the PDAC hybrid clinical trial described in §3 and §4. The documents exist. The IND is written, both protocols are written, the statutory framework is drafted, and the applications are filed. What is absent is money and the institutional instruments money buys. The capitalization plan states the shape of the ask precisely: five Phase I milestones summing to \$306,000 over nine months, then seven Phase II milestones summing to \$1,300,000 over twenty-four, each milestone closed by three concurrent evidence branches that must join before an audit comparison [41]. Seven instruments are absent, and not one can be obtained by the company alone: a qualified site, an operating room, a pathology service, an institutional review board of record, a drug supply agreement, a device acceptance certificate, and a clinical trial agreement.

No further document production changes that. Those seven instruments are the register the whole paper converges on, and none can be produced by writing. The method of §2 solved the problem it was built for; the problem that remains is of a different kind.

9 Conclusions

Between January 2025 and July 2026 the executive branch built the demand side of an AI-assisted cancer research program and left the supply side alone. Executive Order 14179 removed barriers to Federal AI adoption and Executive Order 14212 created the commission that Executive Order 14355 then pointed at cancer [1, 2, 3]. The Childhood Cancer Data Initiative budget doubled to \$100 million on the day that order was signed [6]. The Genesis Mission and its NIH arm committed more than \$1.2 billion [8, 10]. The FDA fielded a generative tool for tasks including clinical protocol review and followed it with an agency-wide agentic platform [12, 13]. CMS built an interoperability framework and a technology office to hold it [14, 15]. On June 22, 2026 HHS launched a department-wide effort to restore American leadership in clinical trials, with Operation TrialBlazer as its oncology arm [16, 17]. Compute, data, agency review capacity, and faster site activation are all being funded. The document a sponsor must file is not, and that is the gap this paper closes on the sponsor side, with artifacts rather than with a proposal.

An IND of twelve modules under 21 CFR §312 and §812, carrying 594,657 characters, 22 figures, 84 section-scoped tables and 52 codified cross-references, was assembled in four days by one author [23]. A Phase 1 protocol at $n = 18$ behind a Phase 0 gate of at least 1000 simulations across two independent frameworks, and a Phase 2 protocol at $n = 220$ across eight high-volume centers targeting a hazard ratio of 0.60, were deposited two days apart and total 369,299 characters and 42 figures [28, 29]. Five bill versions and four companion documents were deposited across nineteen days, ending in a financial data amendment that attaches a cost ledger to every verification run [33, 32, 38]. Fourteen funding artifacts followed, carrying a \$1,606,000 small-business ask against a \$3,500,000 five-year direct program, \$5,900,000 of private capital above a firewall at 3.67 to one, and a virtual trial projected at \$36,330 against an industry benchmark above \$120,000 [41, 39, 31]. Every one of those artifacts was reviewed during development by two model manufacturers independent of the one that produced it, and cleared by a human principal investigator who retained final authority [30, 31].

The two systems are primarily incompatible, and the incompatibility is not a matter of preference. A review regime whose best case is seven to eight weeks and whose typical round is several months cannot absorb an author who deposits more than thirty artifacts in a year, and a review that arrives after release cannot correct the object that was released. A production regime whose unit is the team-month cannot be audited by a commit history it does not keep. A figure format that stores pixels cannot be read back as the input to the next paper. Each of those three is a property of the prior system's architecture rather than of its practitioners, and none is fixable by working harder inside it. The new system's advantages are correspondingly persistent: comprehensive provenance on every artifact, a 1 to 4 day project time scale, AI peer review during development rather than after it, and machine-readable figures that are re-read rather than re-drawn.

What follows is a funding decision, not another document. The author is seeking near-term funding for the PDAC hybrid clinical trial whose IND and protocols are described in §3 and §4, and every artifact named in this paper exists, is deposited under a digital object identifier, and can be read before any money moves. Nothing here is an approved protocol, an active IND, or an enacted statute, and this paper does not claim otherwise: what it claims is that the documents a sponsor must produce can now be produced correctly, comprehensively, and fast enough that the production of them is no longer the rate-limiting step in bringing a pancreatic cancer trial to a patient.

One sentence is worth leaving with a funder. The prior system's rate-limiting step was the document; in the new system it is the institution, and the institution can be bought with money in a way the document could not be bought with effort.

The prior system is not being criticized for lacking a capability it could have had. It is being described as an architecture whose throughput was matched to a production rate that no longer holds on one side of the transaction, and the right response to that is not to defend either side but to say plainly which parts are

now incompatible and to build accordingly. This paper has done that across five main sections, twenty-five figures and twenty-five tables, and every one of the artifacts it reports on is deposited and readable today.

Back Matter

Abbreviations

Every abbreviation this paper uses is expanded below, in two column pairs at the body measure. Trial-specific terms follow the definitions in the published Phase 1 protocol rather than being redefined here.

Short	Expansion	Short	Expansion
AE	Adverse event	IRB	Institutional review board
BICR	Blinded independent central review	ISGPS	International Study Group of Pancreatic Surgery
CCDI	Childhood Cancer Data Initiative	LLM	Large language model
CFR	Code of Federal Regulations	MPR	Major pathologic response
CMC	Chemistry, manufacturing and control	OS	Overall survival
CRO	Contract research organization	PDAC	Pancreatic ductal adenocarcinoma
ctDNA	Circulating tumor DNA	PFS	Progression-free survival
DLT	Dose-limiting toxicity	QSP	Quantitative systems pharmacology
DOI	Digital object identifier	RP2D	Recommended Phase 2 dose
EO	Executive Order	SAE	Serious adverse event
F and A	Facilities and administrative cost	SBIR	Small Business Innovation Research
FDA	Food and Drug Administration	STTR	Small Business Technology Transfer
FTE	Full-time equivalent	USL	Unified Safety Level
HHS	Department of Health and Human Services	VVUQ	Verification, validation and uncertainty quantification
ICH	International Council for Harmonization	3+3	Standard 3 plus 3 dose escalation
CMS	Centers for Medicare and Medicaid Services	ASME	American Society of Mechanical Engineers
HPB	Hepato-pancreato-biliary	RECIST	Response Evaluation Criteria in Solid Tumors
MAHA	Make America Healthy Again	NCI	National Cancer Institute
ARPA-H	Advanced Research Projects Agency for Health	OSTP	Office of Science and Technology Policy
DOE	Department of Energy	NSF	National Science Foundation
FRO	Focused research organization	TikZ	The vector drawing package used for every figure
IDE	Investigational device exemption		
IND	Investigational new drug		

Data and Code Availability

Every work cited to a digital object identifier is openly deposited. The repository is at <https://github.com/kevinkawchak/physical-ai-oncology-trials>, version v4.6.0 [22].

The directory new-trial-system contains the eight-stage sub-prompt schedule, the twenty-five figure specifications across five machine-readable diagram platforms, the three build stages with an Overleaf bundle for each, the master prompt filed verbatim together with the full build output, and the TikZ source of every figure printed here. No raster image is used anywhere in this paper.

Four author-final directories were read directly and are the source of every quotation in §3 through §6: trial-ind/final-ind/publication, trial-protocol/final-protocol/publication, trial-phase-2/final-protocol/publication/author, and funding/capitalization-plan/final-capital/publication. Four input archives were read directly and are the source of every quotation in §5 and §7: the three legislation archives and the AI peer review archive under new-trial-system/inputs. Two further funding directories were read for §6: funding/pdac-funding-applications/final-apply/publication and funding/RFA-RM-27-001-v2.

Author Contributions and Conflicts

Kevin Kawchak is the sole author. He is the sole officer, sole director and sole owner of ChemicalQDevice, which is the sponsor described throughout the documents this paper reports on, and he holds no equity in Revolution Medicines or in any robotic surgery vendor. He is not, and under the plan reported in §6 will not become, a clinical investigator on the trial. No sponsor, contract research organization, institutional review board, regulator, or medical society reviewed this paper or authorized any statement in it. Nothing reported here is an approved protocol, an active IND, or an enacted statute. Claude Code Opus 5 adapted this work; ChatGPT Codex and Google Gemini served as independent reviewers of the work reported in §7, and none of the three is an applicant, investigator, sponsor, regulator, clinician, or decision-maker.

Build Record

This paper was produced by one master prompt in a single turn, decomposed by the model into eight sub-prompts and executed in order, with every generated file committed and pushed the moment it was written. The eight stages produced, in order: six mermaid-type figure specifications, four plantuml-type, six d2-type, four diagrams-type and five graphviz-type, then the draft paper, the full paper, and this final paper, each with a self-contained Overleaf bundle. Each of the three paper stages carries sixteen commits: the style file, the main file, the bibliography, the stage README, eleven section files, a defect-correction pass over the whole stage, and a repository-update commit. The commit history on the working branch is the primary record of that sequence, and it is what the provenance claims of §2 refer to.

Citation

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